From Oil Crisis to Renewable Innovations: Transforming Maritime Policy and Marine Conservation

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Question:
How can renewable marine propulsion technologies and maritime policy reforms work together to reduce the environmental impacts of fossil-fuel-dependent maritime transportation while supporting global trade and coastal communities?

KEYWORDS: Marine Engineering, Combustion Emissions, Marine Governance Systems, Maritime Trade Policies, Sustainable Systems, Renewable Energy Transition , Self-Recharging Engines, Hydrokinetic Energy, Solar Energy, Fossil Fuel.

ABSTRACT:

Marine transportation remains essential to global trade and coastal economies, yet its continued dependence on fossil-fuel propulsion contributes to marine pollution, greenhouse gas emissions, underwater noise, and broader ecological degradation. This research examines how engineering innovation and maritime governance can work together to support a transition toward sustainable marine transportation. First, the environmental effects of conventional two-stroke, four-stroke, and small marine diesel engines are evaluated through their combustion efficiency, pollutant output, and impacts on marine ecosystems. The study then develops its main innovation: a conceptual self-recharging hybrid propulsion system for small coastal vessels that combines photovoltaic energy with retractable hydrokinetic turbines, battery storage, and electric propulsion. Engineering calculations are used to assess energy generation, conversion efficiency, operational models, and the feasibility of the system for an 8-meter artisanal fishing vessel. The research further examines how maritime trade, international regulation, and geopolitical events, particularly the 1973 oil crisis, have shaped global dependence on petroleum and influenced energy policy. Finally, six proposed legal instruments connect renewable propulsion with regulatory compliance, economic incentives, emission transparency, and marine conservation. The findings suggest that reducing the environmental effects of maritime transportation requires more than technological advancement alone. Combining renewable propulsion with effective and accessible regulatory frameworks provides a potential pathway toward cleaner marine ecosystems, greater energy independence, and a more sustainable maritime sector.

INTRODUCTION:

Marine conservation practices have existed for centuries across the world, yet modern industrial activity continues to threaten ocean ecosystems at an unprecedented pace. Geopolitical

upheavals and ecological stress have repeatedly disrupted global trade, placing further strains on the ocean.

Today, the marine industry remains heavily dependent on fossil fuels, with around 90 percent of boat and ship engines powered by them, according to the International Journal of Hydrogen Energy. Heavy fuel oil (HFO) dominates this consumption, powering approximately 72 percent of vessels, followed by marine diesel oil (MDO) at 26 percent, with alternatives such as liquefied natural gas (LNG) accounting for only a negligible share. This dependence comes at a serious environmental cost. Marine vessels emit harmful substances, including nickel, vanadium, oil, and carbon, all of which drive atmospheric forcing, accelerate ocean warming, and acidify marine ecosystems.

Weak governance compounds this damage significantly. A shortage of marine law enforcement has allowed Illegal, Unreported, and Unregulated (IUU) fishing to persist, Marine Protected Areas (MPAs) to go unmonitored, and public awareness to remain low. While the International Maritime Organization has established policies and laws under the MARPOL Convention, enforcement remains inconsistent across many regions. Despite water covering over 70 percent of Earth's surface, comprehensive and strictly enforced marine policies remain largely absent.

This research examines how stronger maritime laws and policies, combined with the adoption of an innovative maritime engineering solution- a self-recharging hybrid propulsion system that harnesses renewable marine energy - can meaningfully improve marine conservation. The central hypothesis is that implementing these eco-friendly propulsion technologies alongside advanced regulatory frameworks will protect marine ecosystems more effectively.

To support this argument, this paper is organized into five key themes, each exploring a distinct aspect of the issue through its own analytical approach. Together, these themes provide a comprehensive framework for aligning marine conservation with sustainable maritime innovation.

Methodology:

This research follows an interdisciplinary and conceptual approach, combining environmental science, engineering, and policy analysis. Each theme was developed using a distinct methodology. Theme 1 relies on a review of peer-reviewed literature on combustion emissions and marine pollution. Theme 2 uses established hydrodynamic equations (Betz limit, kinetic energy flux, drag equation) and validated experimental data from marine turbine studies to design and evaluate the proposed system. Themes 3 and 4 are based on policy analysis and historical review of maritime governance and the 1973 oil crisis. Theme 5 translates these findings into proposed legal instruments, following the structure of existing international

frameworks such as MARPOL and the IMO guidelines. Where data was limited, conservative assumptions were adopted, and all calculations are detailed in the relevant sections.

THEME 1: Ecological Disturbance from Fossil-Fuel Marine Systems:

1.1 – Introduction of Global Impact

Human activities at all scales directly affect marine ecosystems. Recreational boating, particularly when powered by fossil fuels, is often an overlooked source of ecological disturbance. These propulsion systems release harmful pollutants—such as hydrocarbons, nitrogen oxides, carbon monoxide, and particulate matter—that disrupt oceanic habitats in various ways, depending on the engine type and efficiency. All recreational boats powered by petroleum-based fuels, such as gasoline and diesel, pose environmental risks through these impacts. To begin with a broad overview, Table 1 compares two-stroke, four-stroke, and marine diesel engines, common fossil-fuel engines with significant environmental footprints, and their key characteristics.

Table 1: Comparison of small-scale conventional engines and their environmental impact on marine ecosystems

Engine Type	Associated Human Activities	General Operating Efficiency	Dominant Impact	Key Emissions	Severity Indicator
Two-stroke carburetor outboard engine	Frequently in recreational boating and small ocean fishing	Low combustion efficiency due to fuel–oil loss and incomplete combustion during engine operation	Toxicity- Direct release of unburned fuel-oil mixture into aquatic and atmospheric environments due to incomplete combustion (short circuiting)	Hydrocarbons (HC), Nitrogen oxides (NOx), Carbon monoxide (CO), particulate oil residues	Up to 25–30% of fuel may remain unburned, corresponding to hydrocarbon concentrations that can exceed 200,000 ppm under peak operating conditions
Four-stroke carburetor marine engine	Frequently in recreational transport and near-shore vessels	Moderate combustion efficiency with improved fuel regulation, but limited by carburetor-controlled fuel–air mixing	Degradation- Chronic environmental deterioration caused by release of combustion by-products (ethanol) and residual hydrocarbons (petrol and propane)	Hydrocarbons (HC), Nitrogen oxides (NOx), Carbon monoxide (CO)	Approximately up to 9% of fuel may not combust efficiently, with hydrocarbon emissions typically reaching 1,500–5,000 ppm (up to ~10,000 ppm in poorly tuned engines)
Small marine diesel engine	Frequently in small commercial fishing boats and onboard auxiliary power	High combustion efficiency under optimal load, but reduced efficiency under variable conditions due to incomplete combustion	Bioaccumulation- Gradual buildup of diesel fuel contaminants in engine systems and nearby marine conditions due to repeated fuel use and leakage	Nitrogen oxides (NOx), Particulate matter (PM), Sulfur oxides (SOx), low hydrocarbon emissions	Residual fuels can increase NOx emissions by ~20–50% and particulate emissions by ~750–1250%; hydrocarbon emissions remain relatively low (~100–600 ppm, up to ~1000 ppm)

This comparison in Table 1 addresses that environmental impacts are not only a result of fuel type but also of engine design and combustion limitations. The table reveals a close relationship between these effects and thermodynamic limitations, especially in carbureted propulsion systems that mechanically depend on controlled air-fuel mixing. This design reduces

precision under varying temperature and operating conditions, leading to problems of fuel consumption and burning. Such inefficiencies contribute to operational issues such as vapor lock, and their EFI counterparts further increase environmental risks by causing excess emissions (Abdullah). Small marine diesel engines also struggle with thermodynamic constraints related to proper engine loading and precise fuel injection timing. When these engines operate outside their optimal efficiency range, typically between 70 and 85% of rated load, combustion temperatures start to decline, resulting in incomplete fuel burn and the visible release of pollutants such as black smoke (DieselPro). Among conventional marine propulsion systems, thermodynamic inefficiencies are most pronounced in two-stroke carbureted engines, in which their combustion cycle allows a significant portion of unburned fuel-oil mixtures to be poured out into the marine environment.

1.2- Two-Stroke Carburetor Engines

Small recreational boats operating in coastal and oceanic environments rely significantly on two-stroke carburetor engines, which release a disproportionate amount of pollutants relative to their size.

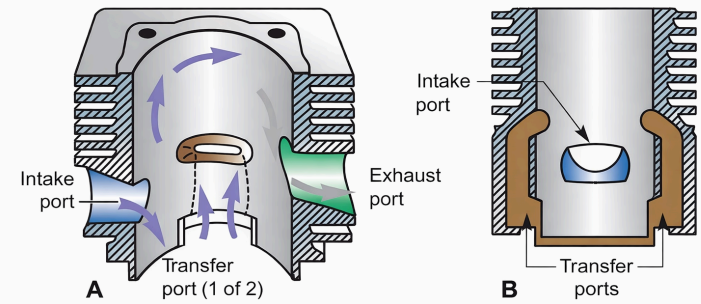


Figure 1: The cutaway cylinder diagram illustrates the positions of the intake, exhaust, and transfer ports within a two-stroke carburetor engine. In the first view, only one transfer port is visible due to the sectioning of the cylinder, while the second view rotates the section to reveal both transfer ports more clearly.

These pollutants stem from the engine's two-stroke cycle, where the piston compresses the fuel-air mixture in the cylinder while simultaneously creating a vacuum in the crankcase. This pressure difference then opens into the reed valve, which allows a mixture of fuel, air, and oil to be drawn in from the carburetor (Marshall Brain). The efficiency of scavenging is critical in a process where incomplete gas exchange directly increases the volume of unburned hydrocarbons released into the surrounding aquatic environment. As the mixture crosses the transfer ports, it is directed from the crankcase into the cylinder above the piston. Inefficient intake port flow causes unburned hydrocarbons and toxic compounds to escape through the exhaust port directly into

marine waters, disrupting shallow aquatic ecosystems that are particularly vulnerable to hypoxia. Recent studies show that optimized intake port designs can improve engine performance, with scavenging efficiency increasing from approximately 74% to 86% under optimized conditions (Liu). These findings highlight how the mechanical limitations of two-stroke carburetor engines contribute to chronic hydrocarbon pollution in marine environments, threatening aquatic biodiversity.

1.3- Four-Stroke Carburetor Engines

Four-stroke carburetor engines are also commonly found in recreational boats, and while they are considered cleaner than two-stroke engines, they still introduce quantifiable pollution that threatens the surrounding biodiversity of marine environments. During the intake stroke, the opening of the intake valve allows a fuel-air mixture to enter the cylinder head as the piston moves down. The mixture is then compressed as the piston rises, increasing pressure within the chamber before heat ignition occurs, generating the combustion force that drives the piston downward again during the power stroke. The exhaust valve finally opens to expel combustion gases into the surrounding marine environment before cycling back to intake (Cameron). In this repeating cycle, their efficiency depends primarily on valve timing, which determines how fresh mixtures replace residual gases within the cylinder. In carbureted systems, fuel-air mixing is mechanically controlled, at which point the valve opens before the piston through the top dead center (TDC) or after the piston through the bottom dead center (BDC). This incomplete combustion releases hydrocarbons and carbon compounds into marine waters, where they accumulate in sediment and disrupt the feeding and reproductive cycles in aquatic organisms (Goodheart-Willcox Co.). While a single recreational boat may seem negligible on its own, the operation of four-stroke engines across popular boating areas creates a chronic source of hydrocarbon pollution in vulnerable ecosystems.

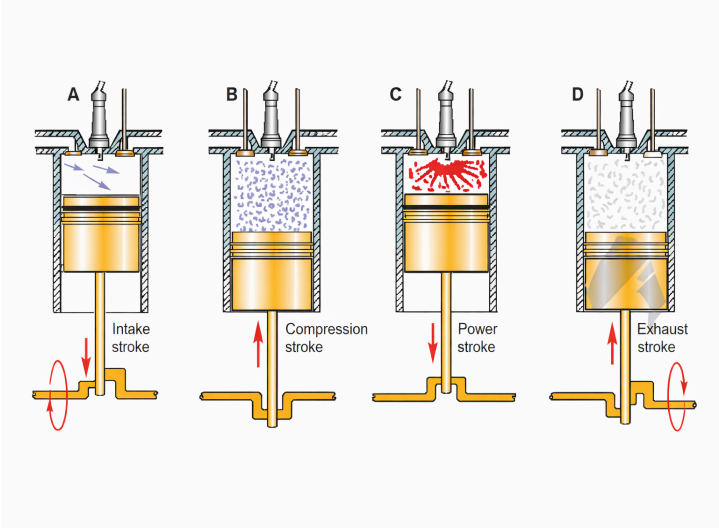


Figure 2: The diagram illustrates the sequence of processes in a four-stroke engine, including intake, compression, power, and exhaust stages. This cycle is completed over two crankshaft revolutions, with only one stroke producing power.

1.4- Small Marine Diesel Engines

Small marine diesel engines are notably used in recreational boating, and despite their reputation for fuel efficiency, their compression-ignition system still releases harmful pollutants that threaten marine ecosystems. As the piston travels upward primarily through a ring, it compresses air at high pressure, forming the lower boundary of the combustion chamber, where ignition occurs. This combustion force is transmitted through the connecting rod, while the piston rings seal the cylinder and manage thermal stress that is generated by pressure (MFM Engineering). Through this process, the efficiency of small marine diesel engines is influenced by compression ratio, which is a way to calculate how effectively air is compressed before fuel injection. Compression ratios typically range from approximately 8:1 to 18:1, which generates pressures of up to 400 to 800 psi within the cylinder. However, weakened piston compression reduces engine performance, releasing smoke and nitrogen oxides into surrounding marine waters, threatening aquatic organisms (Diesel Pro Power). While individual diesel boats may seem minor, their collective operation in heavily trafficked waterways introduces a persistent stream of these toxic chemicals into ecosystems already stressed by other environmental problems.

1.5- Secondary Effects of Thermal & Noise Pollution

(Specific statistics about thermal/noise pollution for readers to notice)

Thermal pollution from recreational boat engines raises local water temperatures by discharging excess heat from engine cooling systems directly into the surrounding environment. When water temperature rises, its capacity to hold dissolved oxygen decreases, leaving aquatic organisms like fish and benthic invertebrates without sufficient oxygen to survive and maintain healthy biological functions. Elevated water temperatures compound this threat by increasing the solubility of toxic compounds such as heavy metals already present in marine waters, meaning pollutants discharged by boat engines become even more chemically dangerous to aquatic life as temperatures rise (Fondriest Environmental Inc).

Beyond thermal disturbances, the noise pollution generated by recreational boat engines across all three combustion systems poses an equally serious threat to marine life, interfering with key functions that many species depend on biologically for survival. Marine mammals such as whales and dolphins rely heavily on sound to communicate and locate prey. But the

continuous noise from boat engines disrupts these essential behaviors, forcing species to alter their calls and change their migration routes. This prolonged exposure to underwater noise can cause trauma, collectively threatening reproductive success and the long-term survival of marine populations already under pressure (Lucy Ellis).

1.6- Segway into Large-Scale Impacts

While these impacts may appear minor at small scales, their cumulative effects exacerbate global marine environmental degradation. When global shipping systems and large fleets utilize these inefficiencies, they accumulate into larger environmental impacts that extend far beyond small-scale marine activity. As maritime transportation continues to be responsible for the majority of global trade, dependence on fossil-fuel emissions is amplifying at a scale that affects entire ocean systems. In response to these large-scale challenges, this paper further aims to develop a water-powered engine as a sustainable alternative that reduces reliance on conventional combustion systems and improves performance across marine industries.

THEME 2 : Harvesting the Ocean's Own Energy: Engineering an Autonomous Hybrid Propulsion System for Small Coastal Vessels

2.1- Introduction: The Environmental Cost of Traditional Marine Propulsion

The ocean has always been at the heart of human development, silently contributing to global trade, transportation, and tourism. However, beyond the efficiency of such a system lies a rising environmental price. Traditional fuel-based propulsion engines continue to disrupt ocean ecosystems, pollute coastal waters, and release harmful emissions into the atmosphere. Therefore, the marine industry is exploring sustainable solutions to reduce these impacts and to find a balance between economic growth and environmental protection. As highlighted in the research paper 'Rebuilding marine life' (2020), while these impacts are severe, restoring the ocean's health is still possible through the adoption of sustainable technologies.

One promising answer is electrification. Researchers identify electric propulsion systems as a sustainable technology solution because they increase energy efficiency and reduce emissions. For example, battery-electric thrusters can cut fuel consumption by 20% in short-range vessels (Smith et al., 2020) and by 35% in 2026 (Global Maritime Forum). Nevertheless, the installation costs remain significantly high, and large-scale implementation remains limited. Moreover, the environmental benefit of electric propulsion depends heavily on the source of the electricity used to charge the batteries. In many parts of the world, this electricity is still generated from fossil fuels such as coal, natural gas, or oil, meaning that the pollution is simply displaced rather than eliminated.

Beyond electrification, renewable energies offer a new perspective that can become a better and lasting solution. While their discussions often focus on solar and wind power, blue energy remains relatively overlooked, even though water covers nearly 70% of the Earth's surface and represents a vast, largely untapped energy resource. Its application in marine propulsion is still underdeveloped, leaving this immense source almost entirely neglected. This situation highlights the urgent need for innovative engineering approaches that can integrate such energy resources directly into marine propulsion systems.

In this context, this theme begins by examining the existing sustainable alternatives in marine propulsion, assessing their potential and limitations, and then focuses on the development of an innovative engineering project: an eco‑friendly, autonomous hybrid propulsion system for small coastal vessels. This system combines two renewable energies—including one blue energy source that remains largely overlooked—into a single integrated architecture. By doing so, it contributes directly to the broader energy transition, offering a practical pathway toward reducing the maritime sector's dependence on fossil fuels. It seeks to advance sustainable maritime engineering by offering a new perspective on harnessing energy directly from the natural marine environment and by proposing an original solution that is not only technically justified but also realistic for small vessels that traditional renewable systems cannot serve.

The reference vessel considered for the development and analysis of this project is an 8-meter artisanal fishing vessel, whose operational profile makes it particularly well suited for demonstrating the feasibility of this autonomous propulsion concept.

2.2 Existing Sustainable Alternatives in Marine Propulsion

2.2.1- Electric Propulsion and Battery-Electric Thrusters:

Nowadays, the most promising way to resolve environmental problems is through electrification. In the maritime context, electrification refers to the transition from fossil-fuel-based combustion engines to electrically powered propulsion systems, where traditional engines are replaced or assisted with electric propulsion, utilizing electric motors powered by batteries or electric power generated through power sources such as fuel cells or generators. A good example of this is battery-electric thrusters, where electric power is converted into mechanical propulsion without burning fossil fuels, resulting in zero direct emissions and a reduction in underwater noise pollution, which is a major problem for marine ecosystems. This transition is also supported by the market, as the Marine Hybrid and Full Electric Propulsion Market Report suggests that the global market is expected to rise from $5.07 billion in 2025 to $5.67 billion in 2026, potentially reaching $8.5 billion by 2030.

From a technical perspective, electric propulsion systems are considered complex and integrated, as they consist of generators, converters, motors, and batteries that require precise energy

control. The hybrid diesel-electric propulsion system is considered an intermediate solution, as it combines the reliability of traditional propulsion systems with the efficiency of electrification systems.

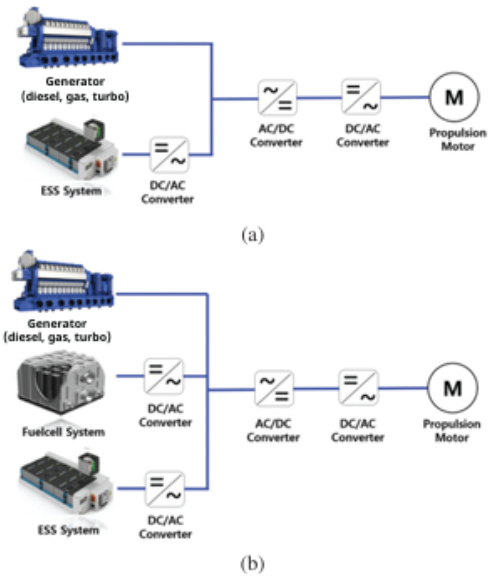


Figure 2.2.1a: Hybrid propulsion, (a) without fuel cell and (b) with fuel cell (Source: N. Arish, 2025)

While Figure 1 illustrates a hybrid propulsion configuration, Figure 2 shows how pure-electric systems can operate either with or without a fuel cell as a power source.

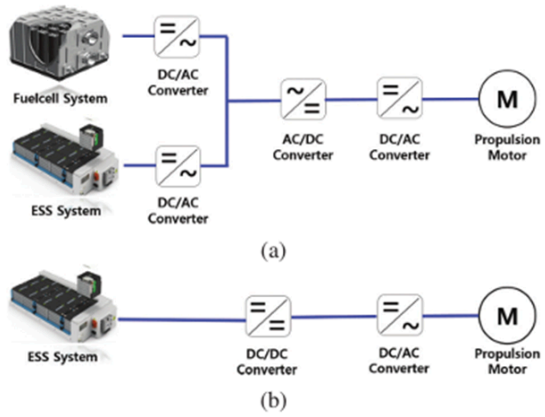


Figure 2.2.1b: Pure-electric propulsion, (a) with fuel cell and (b) without fuel cell (Source: N. Arish, 2025)

However, electric propulsion is not without its challenges. Battery systems remain expensive and limited in capacity, restricting this technology mostly to short-distance vessels. The system also faces critical technical challenges, such as electrical harmonics, overheating, and efficiency losses. Furthermore, if the electricity used for charging still comes from fossil fuels, the

environmental benefit becomes questionable, highlighting that the source of energy matters just as much as the propulsion system itself.

2.2.2 - Solar and Wind Energy in Marine Systems:

As the maritime industry seeks cleaner alternatives to fossil fuels, solar and wind energy have emerged as two of the most accessible and rapidly expanding renewable solutions. Solar energy in marine applications is the conversion of light energy from the sun into electricity by means of the use of photovoltaic panels fixed on the vessel structure or offshore structure, whereas wind energy in marine applications is the conversion of kinetic energy of air by the use of technology such as sail, rotor, and kite sail systems.

These technologies have found large-scale application in offshore energy, as wind speeds are higher and solar panels are cooler due to the effect of the water body. The power produced by a single wind turbine in an offshore application is calculated as:

$$P_{wind} = \tfrac{1}{2} \times \rho \times A_T \times Cp \times U^3 \times \eta g \times \eta gb$$

where ρ is the air density (1.213 kg/m³), A_T is the rotor's swept area (31,400 m² for a 200 m diameter rotor), C_p is the power coefficient (0.32), U is the wind speed, and η_g and η_gb represent the generator and gearbox efficiencies, respectively. Similarly, the power output of an offshore solar PV module is given by

$$P_{solar} = I_{POA} \times A_S \times \eta$$

where IPOA is the plane-of-array solar irradiance, AS is the surface area of the module, and η is the temperature-adjusted conversion efficiency.

In 2023, global offshore wind capacity grew by 10.8 GW, reaching 75 GW worldwide, with China accounting for nearly 60% of new installations. According to the World Energy Outlook, wind and solar combined are projected to meet nearly 70% of global electricity demand by 2050, based on the Net Zero Emissions scenario. Notably, research published in Science Advances (2025) demonstrates that offshore wind and solar PV are naturally complementary since wind energy peaks in winter while solar peaks in summer, providing a more stable and continuous energy supply throughout the year. As seen in Figure 1, exploiting even 1% of the feasible offshore regions would produce around 6,049 TWh from offshore wind, as well as 14,173 TWh from offshore solar, accounting for 21% and 49%, respectively, of global electricity demand.

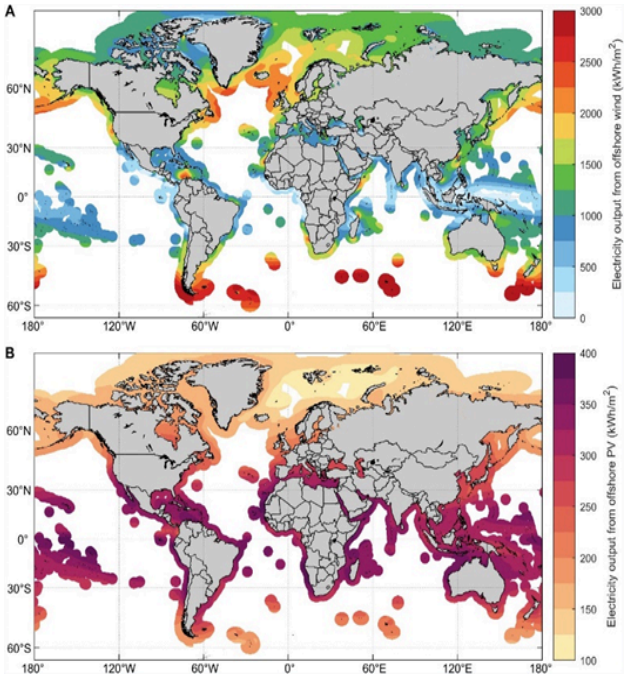


Figure 2.2.2a: Potential electricity output and CO_2 reduction from offshore wind and solar PV (Source: Science Advances, 2025)

At the ship level, solar-powered ships have shown their feasibility in a broad spectrum of uses, ranging from unmanned maritime drones to tourist ships.The Tûranor PlanetSolar , which was the first solar vessel to circumnavigate the globe in 2012 with 537 m² of solar panels providing 93 kW of energy, stands as a notable example of this technology. As per IMO estimates, solar energy usage by ships may result in a 15% reduction in worldwide marine emissions prior to 2050, and the global market for electric ships will expand from $7.25 billion in 2024 to $22.31 billion in 2033.



Figure 2.2.2b – The Tûranor PlanetSolar, the first solar-powered vessel to circumnavigate the globe (2012).

Source: [Wikimedia Commons, https://upload.wikimedia.org/wikipedia/commons/4/41/PlanetSolar-In-Miami-Florida.jpg.](https://upload.wikimedia.org/wikipedia/commons/4/41/PlanetSolar-In-Miami-Florida.jpg)

Solar energy production is limited to sunlight exposure and requires large areas to generate significant power, which is difficult to achieve on most vessels. Offshore installations also need to be resistant to corrosion caused by seawater and harsh environmental conditions, leading to higher expenses. Furthermore, offshore installations are likely to affect marine life, which should be considered during the development phase. Despite these challenges, solar and wind energy remain among the most promising renewable resources for the maritime sector, particularly for large-scale offshore applications and specialized vessels that can accommodate their surface and structural requirements.

2.2.3 - Blue Energy: Sustainable Power from Water

Blue energy refers to all forms of energy gained from the natural movement and properties of water. Covering approximately 361 million km²—nearly 71% of the Earth's surface—the ocean stores an immense and largely untapped energy, reaching a theoretical annual output of 22,848 TWh, far beyond what the world currently consumes. While solar and wind energies are irregular and depend on the weather, most blue energy resources like tidal and ocean currents are highly predictable sources that make a good base for sustainable energy generation systems. This category includes five major types: tidal energy, wave energy, Ocean Thermal Energy Conversion (OTEC), osmotic power, and ocean current energy—each utilizing unique natural properties of water to produce electricity.

Energy extraction from waves relies on wave height and period. Large deep-ocean waves are a significant source of energy, which can last several days. The energy density E (Wh) per unit length and width of the wavefront is determined using the formula:

$$E = (\rho \times g \times H^2 \times T) / 8$$

where ρ is the seawater density (kg/m³), g is gravitational acceleration (m/s²), H is the wave crest height (m), and T is the wave period (s). The power P per unit width of the wavefront is expressed as: $P = (\rho \times g^2 \times Hs^2 \times Te) / (64\pi)$

where H_s is the significant wave height (m) and T_e is the wave period, with an overall efficiency of 30–40%.

Tidal energy is the most technically developed and achieves efficiencies of 80–90%. The La Rance Tidal Power Plant in France, the world's first successful tidal facility, operates with an annual electricity generation exceeding 500 GWh. OTEC uses temperature differences between warm surface water and cold deep water, yet is relatively inefficient with 3–5%. Lastly, the energy produced via the salinity gradient uses the osmotic pressure of different waters (freshwater and salt), providing efficiency at 20–40%.

Among these blue energy sources, ocean current energy, also known as hydrokinetic energy, deserves particular attention. Unlike tidal energy, which depends on the rise and fall of water levels, hydrokinetic energy is extracted directly from the horizontal movement of water currents. These currents are generated by a combination of tidal forces, wind, and differences in water density, and they flow continuously in many coastal regions, straits, and passages around the world. This makes hydrokinetic energy one of the most stable and predictable forms of renewable marine energy.

The power available in a flowing water current is given by the same kinetic energy equation used for wind turbines:

$$P = \frac{1}{2} \times \rho \times A \times v^3 \times Cp$$

where ρ is the seawater density (about 1025 kg/m³), A is the swept area of the turbine rotor, v is the current velocity, and C_p is the power coefficient. This relationship shows that the available power increases with the cube of the current speed, meaning that even moderate currents can carry substantial energy.

Regarding environmental considerations, blue energy does not emit greenhouse gases, and the IRENA estimates that every kilowatt installed can save 1.67 tons of CO_2 each year. Considering the significant expansion of research in the field from 2009 until 2024 and the upcoming trend of hybrid facilities and AI forecasting systems, blue energy becomes a major player among pillars of sustainable maritime engineering.

2.3 Conceptual Design of the Self-Recharging Hybrid Propulsion System

2.3.1 - Concept Overview: An Eco-Friendly, Self-Recharging Marine Propulsion System

Oceans, seas, and coastal waters are home to millions of small working boats, fishing vessels, aquaculture tenders, and harbor craft. Almost all of them rely on diesel engines that are noisy, inefficient, and polluting. These boats are tied to fuel docks, vulnerable to oil prices, and contribute directly to climate change and marine pollution. The need for a clean, autonomous alternative has never been stronger.

The heart of the innovation lies in the complete energy chain that this system introduces. Unlike ordinary electric boats that store grid electricity in a battery, this vessel creates its own energy supply. It captures two renewable resources directly from the marine environment: solar radiation through deck-mounted photovoltaic panels and hydrokinetic energy drawn from natural marine currents by the retractable MERIDYNE turbine. These two sources do not work independently; they are combined into a single central battery that acts as the vessel's power reserve. This battery is the true heart of the system: it stores the harvested energy and then

supplies the permanent magnet electric motor that drives the propeller. The result is a self-recharging propulsion chain in which the boat produces, stores, and consumes its own clean electricity. No fuel tank, no grid connection, no exhaust pipe. The vessel becomes an autonomous power plant, powered by the natural forces of the ocean.

The MERIDYNE turbine is itself a key innovation. Existing hydrokinetic turbines are designed for large tidal farms and cannot fit on small hulls. No such miniaturized, retractable turbine for small coastal vessels exists on the market today. MERIDYNE was therefore developed as a modular family of turbines, available in several rotor diameters so that it can be adapted to boats of different sizes, from light craft under 6 m to trawlers over 12 m. Each model is compact enough to be installed inside the hull and deployed only when the vessel is stationary, capturing energy from the natural current, never from the boat's own motion. This strict separation between propulsion and generation keeps the physics honest and makes the system both efficient and credible.

The working principle is simple but powerful. During the day, photovoltaic panels capture sunlight and send electricity to the battery. When the boat is anchored, drifting, or moored in a current, the MERIDYNE turbines deploy below the hull and extract hydrokinetic energy from the moving water, converting it into additional electrical power. At night, the solar panels rest, but the turbines continue to work as long as the current flows. In port, both sources keep recharging the battery without any human intervention. The result is an autonomous vessel that never needs to return to port for energy.

This concept is aimed at every boat whose operational life includes regular stationary phases: fishing boats, passenger ferries, dive boats, survey vessels, and pleasure yachts. By combining solar and hydrokinetic energy into one integrated architecture, the system offers a genuinely eco-friendly solution for the decarbonization of small-scale maritime transport.

To illustrate this concept, the present study takes the example of an 8-meter artisanal fishing vessel.

The following sections present the detailed working principle, the MERIDYNE turbine, the system design, the technical and engineering analysis, the environmental and economic implications, and the conclusion of this theme.

2.3.2 - Working Principle—Hybrid Integration (Harnessing renewable energies: Hydrokinetic + Solar)

The working principle of the self-recharging hybrid system is simple: the vessel produces, stores, and uses its own electricity. A central battery supplies an electric motor that drives the propeller. This battery is not recharged from the grid; it is recharged continuously by two onboard

renewable sources: a photovoltaic array that captures sunlight whenever it is available and a pair of retractable MERIDYNE hydrokinetic turbines that generate electricity whenever the vessel is stationary or drifting in a marine current.

For the reference 8-meter vessel used throughout this study, the battery capacity is about 60 kWh, the PV array about 2.6 kWp, and each hydrokinetic turbine delivers several hundred watts in a typical coastal current. The working principle remains the same for other vessel sizes, with components scaled accordingly.

The system operates in four modes, summarized in Table 2.3.2.

Table 2.3.2 — System Operational Modes

Mode	Vessel state	Turbines	PV	Battery
1. Navigation	Underway under motor	Retracted	Charging (if daylight)	Net discharging (motor consumes; PV may recharge simultaneously)
2. Stationary in current	Anchored, drifting, or moored in a natural current	Deployed, generating electricity	Charging (if daylight)	Recharging (main mode)
3. Stationary without current	Anchored or moored, no exploitable current	Retracted	Charging (if daylight)	Slow recharge (day) or standby (night)

4. At port	Moored in port or sheltered water	Deployed if port current exists	Charging (if daylight)	Recharging

During navigation, the turbines are retracted inside the hull. The solar panels continue to charge the battery whenever daylight is available, reducing the depth of discharge. When the boat stops in a current, the EMS automatically deploys the turbines, and they begin to produce electricity. At night, the turbines continue to operate as long as the current flows. In port, both sources can recharge the battery without human intervention.

The key idea is temporal complementarity: the two renewable sources are not in competition; they cover different moments. Photovoltaic panels capture energy whenever there is sunlight; turbines capture energy whenever the vessel is stationary or drifting in a current, day or night, which makes the system particularly suited to nocturnal operations. Neither source is asked to do what it cannot: PV panels do not power propulsion alone, and turbines never operate while the vessel is moving under its own motor.

The battery plays two complementary roles. At the daily level, it acts as a temporal buffer between the capture phases (at port, at anchor) and the consumption phases (outbound and return transits). At the multiday level, it constitutes an autonomy reserve that allows the system to absorb unfavorable conditions such as poor weather or weak currents without service interruption. Below a 20% charge threshold, the EMS alerts the operator through the mobile app and automatically restricts nonessential consumption; below 10%, the system signals that an exceptional shore-based recharge is required before the next outing.

The detailed system sizing and the energy balance are presented in Sections 2.3.5, 2.4.1, and 2.4. 2.

2.3.3 - MERIDYNE-70: Retractable Hydrokinetic Turbines for the System

Why a New Turbine?

Existing hydrokinetic turbines such as the Sabella D10 (France, 10 m diameter) and the Marine Current Turbines SeaGen (Northern Ireland, 16 m diameter) are designed for tidal farms and high-flow rivers, with diameters ranging from 2 to 16 meters and weights of several hundred kilograms. These dimensions render them structurally incompatible with a small vessel of 8 to 10 meters. Faced with this lack of a suitable solution, the author of this section proposes an innovation, the MERIDYNE-70 turbine.

The name MERIDYNE combines MER (marine environment, from the French word for sea), DYN (from the Greek *dynamic*, meaning force or power), and E (energy). The number 70 denotes the rotor diameter in cm.

Working Principle

Compact and retractable into the hull, the MERIDYNE-70 works like an underwater wind turbine. When the turbine is deployed, the marine current flows through the swept area and pushes against the three angled blades, creating a pressure difference that generates torque and spins the rotor. This rotation is transmitted directly to the shaft of the PMSG, which converts the mechanical energy into three-phase alternating current.

This raw AC output, however, varies in frequency and voltage depending on current speed, so it is sent to the Central Electrical Cabinet (Section 2.3.5.6) for rectification and stabilization onto the 96 V DC bus. From there, the electricity flows into the central $LiFePO_4$ battery, where it is stored alongside the solar contribution. In short: water current → blade rotation → mechanical torque → electrical generation → storage in the battery, ready to power the propulsion motor.

Power Output
The extractable power from a marine current by a free-flow turbine is governed by the kinetic energy equation for fluids, experimentally validated for marine turbines:

$P = ½ \times \rho \times A \times v^3 \times Cp$

Where:

• $\rho = 1025$ kg/m³ (seawater density)

• $A = 0.3848$ m² (swept area of the 70 cm diameter rotor)

• v = incident current velocity (m/s)

• C□ = 0.42 (power coefficient obtained after the rotor optimization described below)

Power Output as a Function of Current Velocity:

Current (m/s)	**Current (knots)**	**P-hydro per turbine (W)**	**P-elec per turbine (W)**	**P-elec for 2 turbines (W)**
1.0	**1.94**	**83**	**67**	**134**

1.5	**2.92**	**280**	**225**	**450**
2.0	**3.89**	**663**	**532**	**1064**
2.5	**4.86**	**1294**	**1039**	**2078**

Note: The hydrodynamic power values (P_hydro) are calculated with Cp = 0.42. The electrical power actually injected onto the 96 V DC bus (P_elec) accounts for the overall conversion efficiency of the PMSG and power electronics, estimated at 80.3 %. Thus, at 2 m/s, the two turbines together deliver approximately 1064 W of usable electrical power. The complete energy balance, including all conversion stages, is detailed in Section 2.4.2.

Calculation: $P = 0.5 \times 1025 \times 0.3848 \times v^3 \times 0.42$

Under a typical fishing current of 2 m/s (≈ 4 knots), the gross hydrodynamic production of two turbines reaches 6.6 kWh over 5 hours, of which approximately 5.3 kWh is effectively injected into the central battery after conversion losses (efficiency 80.3 %). The complete energy balance is presented in Section 2.4.

Rotor Design
The rotor features three blades made of composite material (fibreglass/reinforced epoxy resin), resistant to saltwater corrosion and impacts with floating debris. The blades follow a NACA 4412 profile, a choice experimentally validated for small-scale marine turbines by Batten et al. (2008). This 4-digit profile (12% thickness-to-chord ratio, 4% camber) offers two decisive advantages:

• **High lift at low speed:** the thick profile generates high starting torque, essential for beginning power production from currents as low as 0.5 m/s.

• **Robustness in real marine conditions:** unlike laminar-flow profiles of the NACA 6-series (e.g., NACA 63-415), which are highly sensitive to water impurities (microalgae, salt, and particulates) that disrupt laminar flow and cause efficiency to collapse, the NACA 4412 maintains its performance across a wide range of surface roughness.

The blade twist (38° root to 7° tip) and elliptical chord (12 cm to 4 cm) were chosen as a design practice for small three-bladed marine turbines at a Tip Speed Ratio of 5–6, aiming to maximize starting torque while limiting tip losses — consistent with the Cp range of 0.40–0.46 reported in

the literature (Bahaj et al., 2007; Batten et al., 2008). A dedicated BEM analysis to numerically validate this geometry remains a priority for future work (Section 2.4.3).

To further increase the energy capture, the blade geometry has been optimized with the following features:

• **Twist:** The pitch angle decreases along the blade following an optimized nonlinear distribution, from approximately 38° at the root to 7° at the tip. This twist ensures that every blade section operates at its optimal angle of attack (6–8°) despite the increasing tangential velocity along the span.

• **Elliptical chord distribution:** The blade is widest near the hub (chord ≈ 12 cm) and tapers smoothly to a narrow tip (chord ≈ 4 cm). This shape captures more energy in the slow inner region and reduces induced drag at the tip.

• **Rounded tip:** The tip is smoothly rounded to mitigate tip vortices without adding complex winglets, keeping manufacturing simple.

• **Surface finish:** A polished gelcoat (roughness < 0.5 µm) with an anti-fouling silicone top coat helps preserve laminar flow and delay biofouling.

An ogival hub reduces flow disturbances upstream of the blades, and the blade roots are smoothly faired into it. The three-bladed configuration offers the best compromise between hydrodynamic efficiency, mechanical robustness, and vibration balancing.

With these optimizations, the expected power coefficient reaches 0.42 at a tip speed ratio of 5–6, yielding about 663 W of hydrodynamic power per turbine at 2 m/s—equivalent to approximately 532 W of electrical power injected into the DC bus. This significantly contributes to the overall energy balance (Section 2.4).

Integrated Generator
The generator is a Permanent Magnet Synchronous Generator (PMSG), housed directly within the turbine's watertight casing and coupled to the shaft in direct drive (without a mechanical gearbox). Advantages:

• Elimination of gearing → reduced mechanical losses, no underwater maintenance, silent operation.

• The rotor starts turning at currents as low as 0.5 m/s (≈ 50 rpm), but usable electrical generation begins at around 0.8 m/s due to the generator's cogging torque.

This gap exists because of cogging torque — the magnetic resistance between the PMSG's magnets and stator teeth, which the rotor must overcome to spin freely. Between 0.5 and 0.8 m/s, the current turns the rotor but too slowly and irregularly to produce a stable, usable voltage. Above 0.8 m/s, the hydrodynamic torque exceeds this resistance, enabling continuous electricity generation.

• Compact assembly (less than 40 cm in axial length), compatible with the 70 cm retraction stroke.

The generator produces three-phase alternating current at variable frequency and voltage, sent to the Central Electrical Cabinet (Section 2.3.5.6) for rectification and stabilization before injection onto the 96 V DC bus.

Retractable Mechanism
Each turbine is placed inside a vertical watertight well cut through the hull. The well is located slightly behind the middle of the boat. The mechanism has three main parts:

• **Electric linear actuator**—This is a waterproof electric jack (IP68 rating). The actuator is powered by the vessel's 24 V auxiliary circuit. This 24 V circuit is supplied by the main 96 V battery through a step-down DC-DC converter. The 24 V standard is safer for submerged equipment and is widely used on boats. When activated, it pushes the turbine down or pulls it up over a distance of 70 cm.

• **Flap valve** – A strong stainless-steel plate (316L) at the bottom of the well. When the turbine is retracted, this flap closes the hole completely. The hull becomes smooth again, so the boat moves without extra drag. A rubber seal around the flap stops water from entering.

• **Guide rails**—Two stainless steel rails inside the well. They keep the turbine straight while it moves up and down, and they absorb the sideways push of the water current when the turbine is deployed.

Thanks to Archimedes' principle, the turbine assembly weighs only about 17 kg in seawater (even though it weighs 25 kg in air). The actuator can easily handle this small load, leaving a big safety margin.

Structural Loading on the Retractable Mechanism

Beyond its submerged weight, the deployed turbine is also subjected to hydrodynamic drag from the current, which the guide rails and actuator must withstand. This force is estimated using the standard drag equation:

$F = ½ \cdot \rho \cdot Cd \cdot A \cdot v^2$

where:

• ρ = 1025 kg/m³ (seawater density)

• Cd = 1.2 (drag coefficient, typical for a disk-shaped rotor)

• A = 0.3848 m² (rotor swept area)

• v = current velocity (m/s)

Calculation: $F = 0.5 \times 1025 \times 1.2 \times 0.3848 \times v^2$

At the reference fishing current of 2 m/s, this yields a drag force of approximately 947 N. At the 4 m/s overspeed threshold that triggers automatic retraction (Section 2.3.3, "Safety Features"), the force rises to about 3 786 N—nearly four times higher, consistent with the v^2 dependency of the drag equation. This sharp increase confirms why the automatic retraction rule is essential: it protects the guide rails and actuator from loads well beyond those encountered during normal operation.

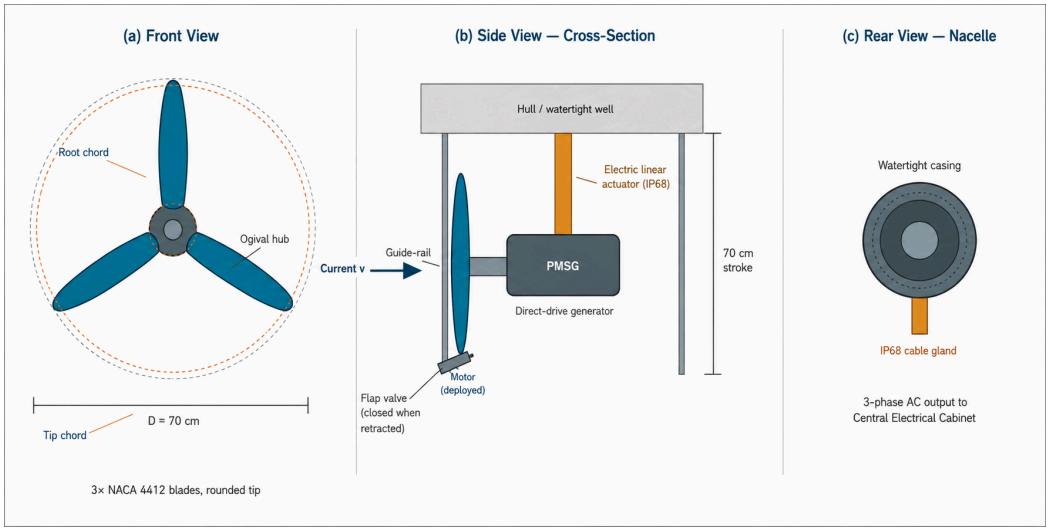


Figure 2.3.3.c — *MERIDYNE-70 turbine design: front, side, and rear views. The rotor is open (no duct), with three NACA 4412 blades of 70 cm diameter. The PMSG is housed in the watertight casing behind the blades.*

Safety Features

The Energy Management System (EMS, Section 2.3.5.6) controls the system according to three safety rules:

• Deployment prohibited if vessel speed > 2 knots (1.03 m/s): excessive hydrodynamic force on the actuator and guide rails.

• Automatic retraction in case of rotor overspeed: current > 4 m/s, risk of blade rupture due to centrifugal force.

• Immediate shutdown in case of abnormal torque: impact with a floating object or rope entanglement around the rotor.

Modular Range
The MERIDYNE turbine is available in several diameters to suit different vessel sizes:

Model	Rotor Diameter	Swept Area	P-elec at 2 m/s	Target Vessel
MERIDYNE-40	40 cm	0.126 m²	174 W	Light craft (< 6 m)
MERIDYNE-70	**70 cm**	**0.385 m²**	**532 W**	**Fishing vessel 8–10 m**
MERIDYNE-100	100 cm	0.785 m²	1086 W	Trawler (> 12 m)

MERIDYNE-70 is the reference model for this paper, as it offers the best compromise between compactness, energy production (1.06 kW with two units), and ease of integration within the hull thickness of an 8-meter vessel.

2.3.4 - The System Design

This section presents the overall system design: the main components, their arrangement on the boat, and their interconnections. The complete architecture is illustrated in Figure 2.3.4.a, which shows a lateral cross-section of the 8-meter reference vessel.



Figure 2.3.4– System overview diagram of the self-recharging hybrid propulsion system for an 8-meter artisanal fishing vessel. The diagram shows the photovoltaic panels, the central battery and electrical cabinet, the PMSM motor, and the two retractable MERIDYNE turbines connected through the 96 V DC bus.

The system combines two renewable energy sources placed directly on the vessel. A photovoltaic array is installed on the cabin roof and bimini to capture sunlight. Two retractable MERIDYNE hydrokinetic turbines are mounted inside the hull, in dedicated watertight wells. They deploy below the hull when the vessel is stationary or drifting in a natural current and retract completely during navigation to avoid drag.

The energy produced by these sources is stored in a central battery bank located low in the hull for stability. The system is managed through a central electrical cabinet that houses the power electronics, the battery management system, the energy management system, and the communication modules. Although the central electrical cabinet (4) is depicted on the exterior cabin wall in the system overview diagram for visual clarity, it is physically housed inside the protected steering cabin in the actual engineering implementation to ensure IP67 watertight integrity and shield the power electronics from marine corrosion. Furthermore, the hydrokinetic turbines (6A/6B) utilize a compact, watertight flush hull-well mechanism to preserve full deck usability during fishing operations.

Propulsion is provided by a permanent-magnet synchronous motor mounted at the stern and connected to the propeller shaft through a marine reduction gear. The crew can monitor the entire system in real time through a wireless interface.

The diagram uses a color code for the power flows: red for solar, blue for turbines, black for battery, orange for general DC distribution, and green dashed lines for communication. This modular design can be scaled for other vessel sizes by adjusting the number of panels and turbines. Detailed technical specifications are provided in Section 2.3. 5.

2.3.5 - System Components and Mechanical Architecture

2.3.5.1 Introduction

This section details all the components of the system, presenting for each one the chosen technology, its main characteristics, and the justification of the design choices. The reference configuration is sized for an 8-meter vessel operating at sea (fishing, service, surveillance), with a modular architecture allowing adaptation to other vessel sizes. All the management and conversion electronics associated with these components are centralized in a single electrical cabinet, detailed in Section 2.3.5.6. The detailed calculations of energy production and consumption are presented in Section 2.4.

2.3.5.2 Photovoltaic Panels

The system uses high-efficiency monocrystalline photovoltaic panels (qualified to IEC 61215 / IEC 61730 standards), chosen for their high efficiency per unit area (18–24%, Fraunhofer ISE, 2025). The main array consists of 6 monocrystalline panels of 400 W each (e.g., SunPower Maxeon 3 400 W, dimensions 1690 × 1046 × 40 mm, $\eta_{module} \approx 22\%$), installed on the cabin roof and bimini top. Two supplementary flexible 100 W panels (18 V/5.5 A, connected in series to form a 36 V string) are mounted on the side decks, adding 200 W of installed power without encroaching on the working area. The total installed peak power is 2.6 kWp, covering a surface of approximately 10.6 m² for the main array plus a small sidedeck area for the flexible panels. The main panels are connected as 2 strings of 3 panels in series ($3 \times 36\ V \approx 108\ V$), and the flexible panels form a separate 36 V string, all feeding the MPPT controller, which stabilizes the output to the 96 V DC bus. The total rigid panel area is about 10.6 m², compatible with the deck space of an 8-meter workboat (detailed sizing and exploitable surface calculations are presented in Section 2.4).



Figure 2.3.5.a*—SunPower Maxeon 400 W monocrystalline photovoltaic panel (1690 × 1046 × 40 mm, 36 V). Source: SunPower / Maxeon (image: empowersolaraustralia.com.au).*

2.3.5.3 Hydrokinetic Turbines (Hydrogenerators)

All the details are in 2.3.3.

2.3.5.4 Battery Storage

The system includes a high-capacity lithium iron phosphate ($LiFePO_4$) battery, chosen for its thermal stability, safety (no risk of thermal runaway), and long lifespan (3,000 to 5,000 cycles, Yang et al., 2019). The battery pack consists of 12 modules of 25.6 V (Victron Lithium NG 25.6 V/200 Ah), connected as 3 parallel strings of 4 modules in series (4 × 25.6 V = 102.4 V nominal), compatible with the central DC bus referenced at 96 V—the nominal industry standard for this power range. The total energy capacity is 12 × 25.6 V × 200 Ah = 61.4 kWh. The capacity is sized to ensure complete weekly autonomy: the vessel accumulates renewable energy during extended periods of rest (port, mooring, anchoring), building a sufficient reserve to cover successive operational outings without external recharging. The core innovation does not lie in the size of this battery but in its recharging mode: powered by onboard renewable sources—photovoltaic panels during daylight hours and hydrogenerators when marine currents are present (at sea, at anchor, but also at port where permanent currents are generally present)—it completely eliminates the need for grid connection at port, which constitutes the main limitation of current marine electric propulsion solutions. A Battery Management System (BMS) continuously monitors each cell's voltage, temperature, and state of charge and protects against overcharge and deep discharge. The battery is positioned in a ventilated compartment at the center of the vessel and as low as possible (in the bilge). This low and centered weight distribution contributes to vessel stability against waves, following the standard principle that lowering the center of gravity reduces roll motion.



Figure 2.3.5.b—*Marine LiFePO₄ battery module from the Victron Lithium NG range (illustration: Victron Lithium NG 25.6V/200Ah; the reference system uses 6 × 51.2V/200Ah modules of the same range). Source: Victron Energy (https://www.victronenergy.com).*

2.3.5.5 Electric Motor (PMSM)

The propulsion system relies on a Permanent Magnet Synchronous Motor (PMSM), sized between 10 and 15 kW for an 8-meter vessel. It operates at an optimal speed of 3000 rpm, reduced to 1000–1500 rpm by a marine gearbox to suit the propeller's hydrodynamic requirements and to enable reverse thrust. Such motors are commercially available and can be ordered without an integrated battery pack. A representative example is the Bellmarine DriveMaster 20.0, a 15 kW continuous (20 kW peak) PMSM designed for a 96 V DC bus, air-cooled and saltwaterproof, and intended for direct connection to the vessel's own battery bank. Being an inboard motor, it drives a conventional propeller shaft; the propeller itself is selected separately based on the hull characteristics.

Technical Specification: Unified Power Hub Configuration

The motor unit is intentionally sourced without a dedicated battery. This configuration is central to the system's energy architecture, establishing a Unified Power Hub:

• **Renewable Inputs**: The photovoltaic array and the 70 cm MERIDYNE hydrokinetic turbines are connected to a single centralized $LiFePO_4$ battery bank, recharging it continuously.

• **Propulsion Power**: The PMSM motor draws its driving power directly from this same central battery bank via a DC→AC three-phase inverter.

• **Energy Recovery:** Under sufficient external currents (e.g., when the vessel is at anchor facing a tidal stream or being towed/under sail), the system can leverage the reversibility of the powertrain. The water flow drives the propulsion propeller, turning the PMSM motor into a generator. This hydro-generation mode injects complementary electrical power back into the central battery bank via the inverter, operating as a secondary energy recovery loop.

By eliminating a dedicated motor battery, the propulsion system, the solar panels, and the hydrokinetic turbines all share the same electrical network and the same storage unit, forming a fully integrated, self-recharging green propulsion loop.



Figure 2.3.5.c – Marine PMSM motor in shaft drive configuration. Example: Bellmarine DriveMaster20.0, 15 kW continuous, 96 V DC, air-cooled, supplied without a proprietary battery. Source: Bellmarine (www.bellmarine.com), a transfluid company.

2.3.5.6 Central Electrical Cabinet

All the management and conversion electronics of the system are centralized in a central electrical cabinet, located in a technical compartment of the cabin. This ventilated marine watertight enclosure groups all electronic subsystems together, simplifying installation, cabling, and maintenance.

The system is designed at 96 V DC, which constitutes the standard in the light marine sector for vessels of this size. This voltage remains within the Extra-Low Voltage (ELV) category (≤120 V DC according to the IEC 61140:2016 standard), ensuring high electrical safety and accessible maintenance by marine technicians, while providing sufficient power for propulsion needs (up to 15–25 kW depending on the motor model).

The cabinet contains:

— The central DC bus, a 96 V distribution rail to which all sources and consumers are connected;

— The power converters: an MPPT controller for the photovoltaic panels, an AC/DC rectifier and a DC/DC converter for each hydrogenerator, a DC-DC step-down converter (96 V to 24 V) for the turbine retraction actuators, and a DC→AC three-phase inverter for the PMSM motor;

— The Battery Management System (BMS), which monitors the battery cells;

— The electronic hub and its embedded software, the Energy Management System (EMS): the EMS monitors in real time the state of each component (voltages, currents, temperatures, and

state of charge), determines the charge and discharge strategy, arbitrates between the renewable sources and the battery, and alerts the user in case of anomaly;

— The communication modules (Bluetooth and WiFi), which transmit system data to a dedicated mobile application, allowing the user to monitor the vessel in real time. This mobile application will be the subject of a dedicated development phase during the prototyping of the system.



Figure 2.3.5.d – Example of a marine central electrical cabinet housing the DC bus, power converters (MPPT, rectifier, DC-DC converter, inverter), BMS, EMS, and communication modules. Such cabinets are commercially available for 96 V DC systems. Source: Walther Electric, Marine & Shipbuilding Power Distribution, www.waltherelectric.com.

2.3.5.7 Cabling and Connections

All external components (photovoltaic panels, hydrogenerators, battery, and motor) are connected to the central electrical cabinet (described in Section 2.3.5.6) by marine cables compliant with the IEC 60092 standard, resistant to salt corrosion, UV exposure, and vibration. This centralized architecture—one dedicated cable per component connecting to the cabinet—simplifies installation and limits total cabling length. The cables linking the hydrogenerators pass through the hull via dedicated IP68-rated watertight cable glands, similar to the motor shaft (watertight stuffing box mentioned in Section 2.3.5.5). Cable cross-sections are sized in accordance with IEC 60092 to handle the high currents characteristic of a 96 V DC system.

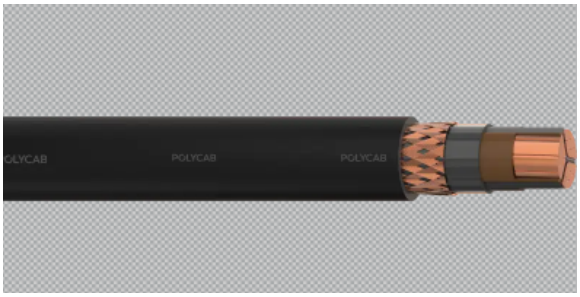


Figure 2.3.5.e – Marine cable compliant with IEC 60092 (tinned copper, UV/salt-resistant sheath). Source: Polycab Polymarine (https://polycab.com).

2.3.6 - Energy Conversion and Propulsion Mechanism

2.3.6.1 Introduction

This section presents the two fundamental processes that allow the system to transform renewable energies into vessel motion: (1) the electrical conversions that adapt the energy of each source to the central DC bus and to the motor, and (2) the propulsion mechanism that transforms the motor's electrical energy into hydrodynamic thrust.

2.3.6.2 Energy Conversion

The system comprises three distinct electrical conversion chains, all centralized in the electrical cabinet (Section 2.3.5.6) and connected to the central 96 V DC bus. This bus is bidirectional: it receives energy from renewable sources and from regenerative recovery and delivers it to the motor and onboard equipment according to the vessel's needs.

The first chain concerns solar energy. The photovoltaic panels produce direct current (DC) whose voltage varies significantly with sunlight, temperature, and shading. A Maximum Power Point Tracker (MPPT) controller continuously adjusts the panel operating point to extract the maximum available power (Zhang et al., 2024) and adapts this variable input voltage to the stabilized 96 V DC bus voltage (typically through a Buck-Boost DC/DC converter with a wide input range).

The second chain concerns the kinetic energy of marine currents captured by the MERIDYNE-70 turbines (Section 2.3.3). The current first sets the blades into rotation (mechanical energy), and this rotation is directly converted into three-phase alternating current by the integrated Permanent Magnet Synchronous Generator (PMSG). The variable-frequency AC current then passes through an AC/DC rectifier, which converts it to direct current, followed by a DC/DC converter that stabilizes the voltage at 96 V before injection onto the central DC bus. The overall efficiency of this hydrokinetic conversion chain (PMSG + rectifier + DC/DC converter) is estimated at 80.3%, as detailed in Section 2.4.2.

The third chain concerns propulsion. The PMSM motor (Bellmarine DriveMaster 20.0, 15 kW) requires three-phase alternating current. A DC→AC three-phase inverter converts the direct current from the DC bus into three-phase AC at variable frequency (controlled by the EMS to adjust motor speed). This variable frequency enables precise speed control from zero to maximum. The inverter also handles controlled reversal of the AC phase order through a deceleration ramp, allowing the motor to be reversed for astern operation (configuration without

a mechanical reversing gear). Marine PMSM architectures with 4-quadrant inverters typically achieve 96% efficiency in propulsion mode (Energies, 2025).

All these conversions are orchestrated by the Energy Management System (EMS), which arbitrates in real time the flows between renewable sources, battery, and motor. The detailed efficiency of each conversion, as well as the overall energy balance of the system, is presented in Section 2.4.2.

2.3.6.3 Propulsion Mechanism

The PMSM motor converts the electrical energy received from the inverter into mechanical rotational energy. This rotation is transmitted to the propeller via a shaft passing through the hull through a watertight stuffing box (Section 2.3.5.5). Depending on the chosen configuration, a marine reduction gear may be interposed between the motor and the propeller to adapt the rotation speed.

The propeller, rotating in the water, transforms this mechanical rotational energy into hydrodynamic thrust. According to Newton's third law, the propeller accelerates a mass of water sternward, creating an equal and opposite reaction that pushes the vessel forward. The blade geometry (profile, pitch, and diameter) is sized to optimize this conversion at the vessel's cruising speed.

Reverse operation is achieved electrically: the inverter, controlled by the EMS, reverses the order of the three-phase AC sequence through a deceleration ramp, changing the motor's rotation direction for astern maneuvers. This configuration eliminates the need for a heavy mechanical reversing gear, reduces maintenance, and protects the electronics from the transient current peaks that would occur during a direct electrical reversal at full speed.

The PMSM motor can also exploit the reversibility of the powertrain to operate as a secondary energy recovery loop. When the vessel is without active propulsion (e.g., at anchor facing a tidal stream or being towed) and subjected to a sufficient water flow, the propulsion propeller is driven into rotation, turning the PMSM motor into a generator. In this hydrogeneration mode, the bidirectional (4-quadrant) inverter rectifies the current produced and reinjects it into the central DC bus (Energies, 2025). This recovery is marginal during simple deceleration (since hydrodynamic drag rapidly slows the vessel) and is only exploitable under the specific conditions described above; moreover, fluid-driven propeller rotation under extreme conditions can induce cavitation, degrading propeller performance (Ocean Engineering, 2024). The efficiency of this recovery is detailed in Section 2.4.2.

Note: the inverter thus integrates two distinct types of reversibility—directional reversibility (reversal of AC phase order for reverse operation) and energy reversibility (reversal of power flow between motor and generator modes).

2.4 Technical and Engineering Analysis

2.4.1 - Energy Resource Assessment and Extraction

This section quantifies the energy that can be harvested from the two renewable sources available on board: marine currents and solar radiation. The calculations use the design parameters established in Sections 2.3.3 and 2.3.5 and provide the input data for the overall energy balance carried out in Section 2.4.2. Equations that represent fundamental physical laws or standard engineering models are accompanied by a reference to their source. Derivations that follow directly from these laws by algebraic manipulation or simple arithmetic do not carry individual citations; their validity rests on the previously referenced formulations.

2.4.1.1 Hydrokinetic Energy Extraction

The MERIDYNE-70 turbine extracts kinetic energy from a freestream water current. The hydrodynamic power available to the rotor is given by the axialflow turbine equation, a standard formulation in marine renewable energy (Bahaj et al., 2007; Batten et al., 2008):

• $P_{hydro} = \frac{1}{2} \cdot \rho \cdot A \cdot v^3 \cdot C_p$ (1)

where:

• $\rho = 1025$ kg/m³ (seawater density),

• $A = \pi \cdot (D/2)^2 = \pi \cdot (0.35)^2 = 0.3848$ m² (rotor swept area for a diameter $D = 0.70$ m),

• v = freestream current velocity (m/s),

• C_p = power coefficient.

To understand the physical origin of the extracted power, one analyses the kinetic energy flux through the rotor disk (Munson et al., 2013). At a current of 2 m/s, the volume of water passing through the swept area each second is

• Volume flow rate = $A \times v = 0.3848$ m² × 2 m/s = 0.7696 m³/s (2)

The corresponding mass flow rate is

• Mass flow rate = $\rho \times$ Volume flow rate = 1025 kg/m³ × 0.7696 m³/s = 789 kg/s (3)

The kinetic energy carried by this mass of water each second is

• Kinetic energy flux = ½ × Mass flow rate × v^2 = ½ × 789 kg/s × (2 m/s)² = 1578 J/s = 1578 W (4)

This 1578 W represents the total kinetic power available in the current passing through the rotor. Only a fraction can be converted into mechanical power; extracting 100% would stop the water, which is physically impossible. The power coefficient C_p quantifies this fraction, and its theoretical upper limit is the Betz limit (59.3%). For a well-designed small turbine, values of 0.4–0.45 are realistic (Bahaj et al., 2007).

The MERIDYNE-70 rotor uses three NACA 4412 blades with nonlinear twist (38° at root to 7° at tip), elliptical chord distribution (12 cm to 4 cm), and rounded tips. This design targets an optimal Tip Speed Ratio (TSR) of 5–6, which is typical for three-bladed horizontal-axis turbines (Bahaj et al., 2007; Batten et al., 2008). According to experimental data on small-scale marine turbines (Bahaj et al., 2007; Batten et al., 2008), a C_p of 0.42 is achievable at these TSR values. Assuming a reference TSR of 5.5, the corresponding rotational speed at 2 m/s is

• $n = (TSR \times v \times 60) / (\pi \times D) \approx (5.5 \times 2 \times 60) / (\pi \times 0.7) \approx 300$ rpm (5)

The mechanical power captured by the rotor is therefore

• P_{hydro} = Kinetic energy flux × C_p = 1578 W × 0.42 = 663 W (6)

This mechanical power is converted into three-phase AC by a direct-drive PMSG, then rectified and stabilized to 96 V DC. The overall conversion efficiency (η) of the generator, rectifier, and DC/DC converter is estimated at 80.3% (losses are broken down in Section 2.4.2). The electrical power injected into the DC bus is

• $P_{elec} = P_{hydro} \times \eta = 663$ W × 0.803 = 532 W (7)

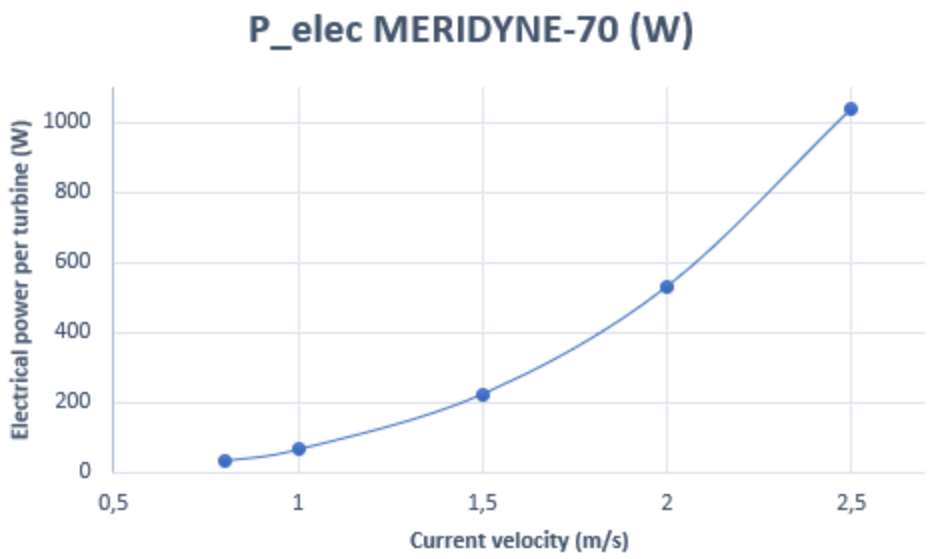


Figure 2.4.1 plots the electrical power of one turbine as a function of current velocity. The cubic relationship implies that power is highly sensitive to current speed; at the reference current of 2 m/s (≈ 4 knots), one MERIDYNE-70 delivers 532 W of continuous electrical power. With two turbines, the system provides 1 064 W.

The complete set of hydrodynamic and electrical power values for current speeds ranging from 1.0 to 2.5 m/s is provided in the detailed turbine specifications (Section 2.3.3, Table 2.3.3). All power values are instantaneous (1 W = 1 J/s). As a concrete illustration, during a typical 5-hour stationary fishing session at 2 m/s, one turbine would deliver 2.66 kWh of electrical energy, and two turbines would deliver 5.32 kWh. The integration over time for any type of mission is performed in the daily energy balance (Section 2.4.2).

2.4.1.2 Solar Energy Extraction

Solar energy is harvested through the photovoltaic (PV) effect. When sunlight strikes a monocrystalline silicon cell, photons with energy greater than the bandgap (1.12 eV for silicon) are absorbed, generating electron-hole pairs. An internal electric field at the pn junction separates these charges, creating a direct current (Smets et al., 2016). In this way, the incoming solar irradiance (energy flux in W/m²) is directly converted into DC electricity by the cell.

The instantaneous electrical power produced by a PV module depends on the incident solar irradiance (G, in W/m²) and the module efficiency (η_module). Under standard test conditions (STC: G_STC = 1 000 W/m², cell temperature 25 °C), a module of area A_module delivers (Masters, 2013):

• $P_{module,STC} = A_{module} \times G_{STC} \times \eta_{module}$ (8)

For the 8 m workboat, the main array consists of 6 high-efficiency monocrystalline panels of 400 W each (e.g., SunPower Maxeon 3 400 W, dimensions 1 690 × 1 046 × 40 mm, $\eta_{module} \approx 22\%$, Fraunhofer ISE, 2025; IEC 61215/61730 qualified), installed on the cabin roof and bimini top. Two supplementary flexible 100 W panels (typical 18 V/5.5 A, connected in series to form a 36 V string) are mounted on the side decks (gunwales), adding 0.2 kWp without encroaching on the working area. The total installed peak power is

• $P_{peak} = 6 \times 400\ W + 2 \times 100\ W = 2.6\ kWp$ (9)

The panels are connected as 2 strings of 3 panels in series for the main array ($3 \times 36\ V \approx 108\ V$) and one additional 36 V string for the flexible panels, all feeding the MPPT controller which stabilises the output to the 96 V DC bus (Section 2.3.5.2).

To estimate the daily energy production, the concept of peak sun hours (PSH) is used (Duffie & Beckman, 2013). PSH represents the number of hours per day during which the sun would deliver a constant 1 kW/m² to the panels. For temperate and tropical coastal regions, the average daily global horizontal irradiance (GHI) is about 5 kWh/m², which corresponds to 5 PSH. In tropical regions, this value can reach 6–7 PSH (NREL, 2023).

The daily DC energy generated before system losses is therefore

• $E_{dc} = P_{peak} \times PSH = 2.6\ kW \times 5\ h = 13\ kWh/day$ (10)

However, not all of this energy reaches the DC bus. Several loss factors must be accounted for (Marion et al., 2005; Dobos, 2014):

• MPPT efficiency (~98%): the tracker does not find the exact maximum power point at all times.

• Wiring and connection losses (~98%): resistive losses in the cables.

• Temperature derating (~92%): panels heat up above 25 °C, reducing their voltage and efficiency.

• Soiling, shading, and mismatch (~95 %): dust, bird droppings, partial shading, and slight differences between modules.

The overall system efficiency from panel to 96 V DC bus is

• $\eta_{sys} = 0.98 \times 0.98 \times 0.92 \times 0.95 \approx 0.84$ (84%) (11)

The net daily energy injected into the DC bus is

• $E_{solar} = E_{dc} \times \eta_{sys} = 13\ kWh \times 0.84 \approx 10.9\ kWh/day$ (12)

This value represents the solar contribution alone; the hydrokinetic contribution was quantified in Section 2.4.1.1 and both are combined in the overall energy balance (Section 2.4.2). Taking into account day to day variability and occasional cloud cover, a conservative value of 10–11 kWh/day is adopted for the energy balance in Section 2.4.2. The MPPT controller continuously adjusts the panel voltage to maintain operation at the maximum power point, feeding the 96 V DC bus.

2.4.2 - Energy Efficiency Evaluation

This section evaluates the propulsion chain efficiency and presents the overall energy balance of the self-recharging hybrid system. The calculations are illustrated with the example of a typical 8-meter artisanal fishing vessel, which is representative of the Mediterranean and Atlantic coastal fleet. According to FAO (2022), such boats routinely undertake daily trips lasting 10–12 hours, often including nighttime operations (e.g., lamp fishing, set nets, and longlines), during which the hydrokinetic turbines can continue to produce electricity. The reference scenario—5 hours of daytime fishing, 3 hours of overnight drift (8 hours total fishing time), and 3 hours of transit—therefore reflects real-world operating patterns. For larger vessels (10–12 m), the same concept can be applied by upgrading to MERIDYNE-100 turbines and increasing the photovoltaic area, further improving the energy balance.

Propulsion chain efficiency

The propulsion chain converts the battery's stored energy into thrust. Its overall efficiency is the product of the successive stage efficiencies, taken from the marine engineering literature and manufacturer data:

• $LiFePO_4$ battery (charge/discharge): $\eta_{bat} = 95\ \%$ (Yang et al., 2019)

• DC→AC three-phase inverter: $\eta_{inv} = 96\ \%$ (Section 2.3.6; Energies, 2025)

• PMSM motor (Bellmarine DriveMaster 20.0, 15 kW): $\eta_{mot} = 94\%$ (Applied System Innovation, 2025). *Note: the 96% efficiency cited in Section 2.3.6 for "marine PMSM architectures with 4-quadrant inverters" refers to the combined motor+inverter efficiency. The two stages are separated here for a finer breakdown, hence the conservative 94% for the motor alone.*

• Marine reduction gear: $\eta_{red} = 97\%$ (standard value; Molland et al., 2011)

• Propeller (mechanical → thrust): η_{hel} = 70% (Molland et al., 2011)

Overall battery-to-thrust efficiency:

$\eta_{prop} = 0.95 \times 0.96 \times 0.94 \times 0.97 \times 0.70 \approx 0.58$ (58%)

Excluding the propeller (whose efficiency varies with hull shape and vessel speed), the battery-to-shaft efficiency is

$\eta_{shaft} = 0.95 \times 0.96 \times 0.94 \times 0.97 \approx 0.83$ (83 %)

This shaft efficiency is comparable to standard grid-charged electric propulsion (80–90 %) and about two to three times higher than marine diesel engines (30–40 %) or gasoline outboards (25–30 %) (Heywood, 2018). A detailed comparison is provided in Section 2.4.4.

The decisive advantage is the energy source: the battery is recharged by onboard solar and hydrokinetic sources, eliminating shoreside grid electricity and making the vessel fully autonomous.

Typical day energy balance

Two characteristic daily situations are considered: an active sea day (fishing trip with overnight drift) and a port day. The vessel is assumed to be equipped with 6 monocrystalline 400 W panels plus 2 flexible 100 W panels (2.6 kWp total, see Section 2.4.1.2) and to cruise at an economic speed of 4–5 knots, requiring about 8 kW of propulsive power.

Active sea day (with overnight drift)

• Outward transit (1.5 h, 8 kW propulsion + 0.5 kW auxiliaries in navigation mode): 12.75 kWh

• Return transit (1.5 h, 8 kW propulsion + 0.5 kW auxiliaries in navigation mode): 12.75 kWh

• Daytime fishing (5 h, vessel stationary or drifting, turbines deployed, auxiliaries active at 0.5 kW): 2.50 kWh

• Nighttime drift (3 h, turbines deployed, auxiliaries in reduced mode at 0.3 kW): 0.90 kWh

• **Total consumption: 28.90 kWh**

• Solar production (daytime, 5 PSH): 10.90 kWh

• Hydrokinetic production (2 MERIDYNE-70 turbines, deployed during the 8 h of stationary phases—5 h daytime fishing + 3 h nighttime drift—average current 2.0 m/s): 2 × 532 W × 8 h = 8.51 kWh

• **Total production: 19.41 kWh**

Net deficit at sea: 28.90 − 19.41 = 9.49 kWh (self-sufficiency ≈ 67%)

Port day

• Auxiliary consumption (EMS/BMS in low-power standby, ~20 W): 0.50 kWh

• Solar production: 10.90 kWh

• Hydrokinetic production (port current 1.0 m/s, 2 turbines × 67 W × 24 h): 3.22 kWh

• **Net surplus at port: (10.90 + 3.22) − 0.50 = 13.62 kWh**

Equilibrium over several days

One port day compensates for about 1.4 times the deficit of one sea day (13.62 / 9.49 ≈ 1.44). An alternating rhythm of one sea day and one port day therefore keeps the battery balanced with a comfortable margin. Several consecutive sea days draw on the 61.4 kWh battery, which can supply the deficit for more than six days without any port recharge. As soon as the vessel stays a few days in port (bad weather, weekends, maintenance), the battery is fully replenished. The BMS and EMS prevent overcharging.

Sensitivity and robustness

In sunnier regions (6–7 PSH), solar production rises, further reducing the deficit. In areas with strong tidal currents (≥2.5 m/s), hydrokinetic power more than doubles, and the system easily becomes surplus at sea. Conversely, very intensive weeks draw on the battery reserve, which is restored upon return to port. The modular design also allows the integration of additional flexible panels for extra energy security. As a further development perspective, the addition of a small wind turbine could be considered for regions with consistent winds.

Overall, the self-recharging hybrid system covers about two-thirds of its at-sea energy needs directly from onboard renewable sources and achieves a comfortable energy equilibrium through the natural alternation of active and idle days, without imposing a fixed schedule. The battery acts as a buffer, guaranteeing complete autonomy without shoreside infrastructure.

2.4.3 - Limitations and Future Developments

The self-recharging hybrid propulsion system has been designed to maximize energy autonomy for small coastal workboats, but it is subject to several inherent limitations that must be acknowledged. At the same time, several promising avenues exist for further improving the system's performance and extending its applicability.

Dependence on environmental conditions

• **Current velocity**: the MERIDYNE-70 turbines begin producing useful electrical power from approximately 0.8 m/s (Section 2.3.3). In regions where the current remains consistently below this threshold, the hydrokinetic contribution drops significantly, and the energy deficit must be covered by solar power alone or by the battery.

• **Solar irradiance**: daily solar production varies with latitude, season, and weather. The reference value of 10.9 kWh/day assumes 5 peak sun hours, which is typical of temperate and Mediterranean coastal zones. In high-latitude winters or during extended cloudy periods, this figure can be halved, reducing the atsea self-sufficiency below 50%.

• **Nighttime operation**: the overnight hydrokinetic gain relies on the vessel remaining stationary in a current for several hours. If the fishing schedule does not include nighttime drift or if the skipper returns to port every evening, the ~3.2 kWh of extra hydro production is lost, and the daily deficit increases.

Initial investment cost

Although operational costs are zero, the upfront cost of the system—six high-efficiency solar panels, two MERIDYNE-70 turbines with their retractable mechanisms, a 61.4 kWh $LiFePO_4$ battery, the PMSM motor, and the power electronics—is higher than that of a conventional diesel outboard or a basic electric drive. A detailed economic payback analysis, which falls outside the scope of the present study, would be necessary to determine the breakeven point for different usage profiles.

Maintenance of submerged mechanical parts

The retractable turbine mechanism (linear actuator, flap valve, and guide rails) operates in a harsh marine environment. Although designed with IP68-rated components and corrosion-resistant materials, the seals and moving parts will require periodic inspection and possible replacement, adding a maintenance task that does not exist on a pure electric or diesel boat.

Battery aging

$LiFePO_4$ batteries have a long cycle life (3 000–5 000 cycles, Yang et al., 2019), but their capacity gradually degrades. After 10–15 years of regular use, the usable capacity may drop below the 61.4 kWh design value, slightly reducing the buffer autonomy. This effect can be compensated by oversizing the battery at installation or by planning a midlife replacement.

Geographical applicability

The system is optimized for coastal regions with moderate to strong tidal or permanent currents (e.g., Mediterranean straits, Atlantic shelf, Indonesian Throughflow). In purely wind-driven or very calm seas, the hydrokinetic resource may be insufficient year-round, limiting the system's attractiveness to specific geographical niches.

Future developments

Several technological improvements could further enhance the performance of the self-recharging hybrid system without altering its core architecture. Bifacial solar panels, once available in the required dimensions, could provide a 5–15% gain in daily solar production depending on surface albedo (NREL, 2023). A small marine wind turbine (100–200 W, e.g., Rutland 504 or Silentwind Pro) could be added to the modular DC bus to supply complementary energy during overcast or nighttime periods, particularly in consistently windy regions. The Energy Management System could be upgraded with predictive algorithms that use weather forecasts and tidal models to optimize the charge-discharge cycles of the battery and extend its lifetime. In the longer term, emerging battery technologies such as solid-state or sodium-ion cells could offer higher energy density or lower cost, further improving the system's economic viability.

Despite these limitations, the self-recharging hybrid concept demonstrates that a well-dimensioned combination of solar and hydrokinetic energy can cover the majority of a small workboat's daily energy needs. The following comparative analysis (Section 2.4.4) shows that, even with these acknowledged constraints, the system outperforms both diesel and standard electric alternatives on most operational criteria.

2.4.4 - Comparative Analysis: Self-Recharging Hybrid vs. Standard Electric vs. Diesel

This section compares the proposed self-recharging hybrid system with the two main alternatives for an 8-meter artisanal fishing vessel: a conventional diesel engine and a standard grid-charged electric drive. The comparison considers complete propulsion systems, including the energy source, storage, conversion, and thrust components, not just the prime mover alone. The reference mission follows the realistic daily profile established in Section 2.4.2 (3 h of transit at 8

kW, 5 h of daytime stationary fishing, 3 h of overnight drift), which is typical of Mediterranean and Atlantic small-scale fisheries (FAO, 2022).

Although the self-recharging hybrid uses an electric motor (PMSM), it is fundamentally different from a standard electric boat: it does not depend on the grid to recharge. Its energy is harvested on board from solar radiation and marine currents, stored in a $LiFePO_4$ battery, and used directly for propulsion and auxiliaries. The comparison below distinguishes between the motor technology (electric in both cases) and the energy supply chain (autonomous vs. grid-dependent).

Table 2.4.4 – Comparative analysis of propulsion systems for an 8-meter coastal workboat.

Criterion	Self-recharging hybrid	Standard electric	Diesel	Source
Energy source	Solar + hydrokinetic (free)	Grid electricity	Diesel fuel	–
Efficiency (source → shaft)	83 % (battery-to-shaft)	80–90 % (battery-to-shaft)	30–40 % (tank-to-shaft)	Heywood (2018); Applied System Innovation (2025)
Daily energy consumption	28.9 kWh (self-supplied)	28.9 kWh (from grid)†	~7–10 L diesel (~70–100 kWh thermal)‡	Section 2.4.2; FAO (2022)

Daily energy cost (€)*	**0**	**~5.78**	**~12–18**	**European coastal prices august 2026** **https://www.fuel-prices.eu/**
Annual energy cost (200 sea days)	**0 €**	**~1 156 €**	**~2 400–3 600 €**	**–**
Autonomy at sea	**Effectively unlimited – the battery recharges whenever the vessel is in the sea (fishing, overnight drift); no need to return to port for energy**	**Typically 1–2 days (with a 40–60 kWh battery); once depleted, the boat must return to port to plug in**	**5–10 days (fuel tank); once empty, must refuel at a dock**	**Section 2.4.2; manufacturer data (Torqeedo, Oceanvolt, Bellmarine)**
Energy dependence	**Depends on climate (solar irradiance + current velocity); no**	**Depends on grid electricity availability and price**	**Depends on diesel fuel availability and price**	**–**

	external supply needed			
Direct CO_2 emissions (kg/day)	**0**	**0 (indirect, depends on grid mix)**	**~19–27 (based on 7–10 L × 2.68 kg CO_2/L)**	**ADEME (2023)**
Maintenance	**Preventive, annual (turbine seals, panels, electrical connections)**	**Low (motor + battery)**	**High (oil, filters, injectors every 100–200 h)**	**Molland et al. (2011); manufacturer data**
Infrastructure required	**None**	**Shore power point**	**Fuel dock**	**–**

–

† For an equivalent mission profile and propulsion efficiency.
‡ For the same mechanical work output, assuming a diesel engine efficiency of ~35 % (Heywood, 2018).

• Representative European coastal market prices (~0.20 €/kWh, ~1.80 €/L diesel); actual values vary significantly by region.

What makes the hybrid fundamentally different?

A standard electric boat of this size typically carries a 40–60 kWh battery (e.g., Torqeedo Deep Blue, Oceanvolt AXC, Bellmarine DriveMaster). With a daily consumption of 28.9 kWh, this gives an autonomy of roughly 1.4 to 2.1 days—enough for a single fishing trip, but not for a

multiday mission. Once the battery is depleted, the boat must return to port and connect to shore power. The diesel boat faces the same constraint with its fuel tank.

The self-recharging hybrid carries a battery of similar capacity (61.4 kWh), but it **refills its battery continuously while at sea**. The MERIDYNE turbines generate electricity from the current whenever the boat is stationary—precisely during the long hours of fishing and the overnight rest. The solar panels contribute additional energy throughout the day, even while navigating. As a result, the battery never reaches zero, and the boat never *needs* to return to port for energy. It only goes back when the crew decides to, for rest, supplies, or to sell the catch. This is the fundamental innovation: not a larger battery, but a battery that recharges itself during the normal pauses of every fishing trip, completely eliminating the dependence on shoreside infrastructure.

Economic considerations

The initial purchase cost of the self-recharging hybrid is higher than that of a basic diesel engine or a simple electric outboard because it includes the photovoltaic array, the MERIDYNE turbines with their retractable mechanisms, and the larger $LiFePO_4$ battery. However, the daily energy cost is zero, and the maintenance is far simpler and cheaper than that of a diesel engine. Over a typical vessel lifetime of 15–20 years *(FAO, 2022)*, the savings on fuel or electricity alone offset the higher upfront investment, making the hybrid the most economical solution in the long run.

The self-recharging hybrid is the only architecture that combines the high efficiency and low mechanical complexity of an electric drive with the unlimited autonomy and independence of a diesel boat while producing zero direct emissions. It frees small-scale fishers from the constraints of fuel docks and electrical outlets and provides a credible, sustainable alternative for coastal operations worldwide.

2.5 - Environmental and Economic Implications

<u>The environmental burden of conventional marine engines</u>

A destructive chain that begins with fuel

Every liter of diesel or gasoline burned in a marine engine sets off a cascade of damage. It emits CO_2, nitrogen oxides, and fine particles that directly fuel global warming and degrade coastal air quality. Simultaneously, fuel and oil leaks spread a toxic film across the water surface, blocking oxygen exchange, poisoning plankton and fish, and depositing heavy metals—lead, mercury, and cadmium—into the food chain (J. Mar. Sci. Eng., 2020). These pollutants contaminate seafood and can cause serious health effects in humans, including neurological damage and developmental problems in children. Engine noise and waste heat further disrupt marine habitats

and stress the species that coastal communities and ecosystems depend on. Whether the vessel is a fishing boat, a passenger ferry, a dive boat, or a leisure craft, the environmental consequences are the same: every hour of engine operation leaves a lasting imprint on the water and the air.

From the ocean to the land: climate knows no borders.

The CO_2 released at sea does not stay there. It accumulates in the atmosphere and drives a global rise in temperature. On land, this translates into prolonged droughts, desertification, collapsing crop yields, and massive livestock losses. When the land can no longer feed its people, they are forced to move, fuelling migration, food insecurity, and political instability across entire regions. A problem that starts with a boat engine becomes a planetary crisis affecting every continent.

The geopolitical price of fossil fuels

Dependence on petroleum is not just an environmental risk. It is a strategic vulnerability. Nations compete and sometimes go to war over control of oil and gas reserves. Trade routes are militarized, and the wealth from fossil fuels often entrenches corruption instead of building sustainable economies. For any vessel operator—whether a small-scale fisher, a charter captain, or a ferry owner—this global game means unpredictable fuel prices and supply chains that can break overnight. Energy independence is therefore not a luxury: it is the only way to protect maritime livelihoods from decisions made in distant capitals and oil markets.

The ticking clock of non-renewable resources

Beyond the pollution they cause, fossil fuels carry a more fundamental flaw: they are finite. Every barrel of oil and every cubic meter of gas burned today is gone forever. The reserves that power the global maritime fleet are not replenished. As easily accessible deposits shrink, extraction becomes more expensive, more environmentally destructive, and more likely to trigger conflicts over what remains. Eventually, the resource will simply run out or become too costly for ordinary people to afford. A propulsion system that depends on an exhaustible fuel has no long-term future. Only systems that run on permanently available sources—the sun, the wind, the currents—can outlive the fossil fuel era.

Why a standard electric boat is not the answer

A conventional electric boat may be quiet and emission-free at sea, but its battery is charged from a shore-side grid that, in many parts of the world, still runs on coal, natural gas, or heavy fuel oil. The pollution is simply moved from the exhaust pipe to the power plant chimney. It is not eliminated. A truly clean propulsion system must produce its own energy on board, without leaning on a fossil-dependent grid.

The scale of the problem

The figures are staggering. In 2023, the global maritime sector consumed approximately 210 million tonnes of fuel, or more than 200 billion liters (IMO, 2024). Diesel and gas oil represent roughly 9% of that total, nearly 19 million tonnes burned in a single year. Zooming in, a single coastal trawler, passenger boat, or workboat can swallow 8 000 to 10 000 liters of diesel every week, adding up to 400 000 to 500 000 liters annually. These numbers leave no doubt: every vessel that switches to renewable propulsion makes a measurable difference.

How the self-recharging hybrid system reverses the chain

The proposed system cuts the problem at its root. Solar panels and MERIDYNE hydrokinetic turbines harvest energy directly from the environment: no combustion, no fuel, no grid connection. The PMSM electric motor operates in silence, eliminates underwater noise, and releases zero emissions into the air or water. By replacing diesel with self-generated renewable electricity, the system removes the first domino in a long chain of environmental, health, and economic damage. On land, it helps slow the climate-driven degradation of agriculture and water resources. At sea, it keeps the water clean, the fish healthy, and the seafood safe—regardless of whether the boat is used for fishing, transport, tourism, or leisure.

Economic sense for vessel operators

Operational costs drop to zero: no more fuel bills, no more oil changes, and no more exhaust repairs. A typical coastal boat saves around €12–18 per day (based on 7–10 L at €1.80/L), i.e., €2 400–3 600 per year for a 200-day operating season. Over the 15- to 20-year life of the vessel, those savings far outweigh the higher upfront investment. Furthermore, operators are no longer at the mercy of volatile oil prices or the geopolitical shocks that disrupt supply.

Autonomy and dignity

Energy independence transforms the daily reality of anyone who works or travels on the water. Boats are no longer tied to fuel docks or charging points; they can stay at sea as long as the mission requires. The absence of exhaust fumes, vibrations, and engine roar makes the deck a healthier, safer place. And because the technology can be installed and maintained by local technicians, it creates skilled jobs and strengthens the local economy rather than draining it.

A solution for every vessel that waits at sea

The self-recharging hybrid system is not limited to fishing boats. In this study, an artisanal fishing vessel was chosen as a concrete example because its operational profile—long hours at sea, regular stationary phases for work and rest, and a typical alternation between active days and

port days—makes it an ideal case to demonstrate the system's capabilities. However, the underlying principle is universal: whenever a vessel stops, it recharges. Passenger ferries waiting at a dock, dive boats drifting over a reef, survey vessels holding station, or even pleasure yachts anchored in a bay all spend hours in conditions that are ideal for harvesting solar and hydrokinetic energy. Any vessel whose operational profile includes regular stationary periods can therefore achieve the same energy autonomy demonstrated in this study, regardless of its primary activity.

2.6 - Conclusion of the Theme

This theme introduced an original self-recharging hybrid propulsion system designed to free coastal vessels from fossil fuels and shore-side electricity. Its core innovation rests on two pillars. The first is the MERIDYNE hydrokinetic turbine, a compact, retractable unit developed specifically to harvest energy from low-speed marine currents on small boats, something no existing turbine on the market can do. The second is the intelligent integration of this turbine with solar photovoltaic panels into a single unified electric architecture. Together, they feed a central battery that powers the electric motor, enabling the vessel to recharge continuously while stationary, navigate without fuel, and remain at sea indefinitely without ever needing a port connection or a fuel dock.

The engineering analysis confirmed the technical viability of this concept. The propulsion chain achieves a battery-to-shaft efficiency of 83%, matching the best electric drives available. In a realistic daily scenario based on real-world operating patterns, the onboard solar and hydrokinetic sources cover 67% of the vessel's energy consumption at sea. The remaining deficit is drawn from a 61.4 kWh $LiFePO_4$ battery, which alone provides more than six days of full autonomy. Because the MERIDYNE turbines recharge the battery whenever the boat is stationary during work, during rest, and while docked in a current, the system never depends on returning to port for energy. The natural alternation of active days and idle days makes it effectively unlimited.

This work reveals the true power of renewable energies in the maritime world. Sunlight and ocean currents are not merely supplementary sources; they are permanent, inexhaustible forces that, when harnessed intelligently, can drive real vessels in real conditions. By capturing these energies directly on board, the system eliminates the need for diesel entirely in coastal operations, demonstrating that the global energy transition is not a distant goal but a present possibility. It shows that a cleaner, quieter, and more equitable maritime future is already within reach, powered by the natural rhythms of the planet itself.

The concept is scalable and designed for the future. Although developed and dimensioned around an 8-meter artisanal fishing vessel, the same architecture applies to any boat that spends time stationary at sea: passenger ferries, dive boats, survey vessels, or pleasure yachts. Larger vessels can adopt MERIDYNE-100 turbines and expanded solar arrays without changing the underlying principle. Looking ahead, the integration of bifacial panels, small marine wind turbines, predictive energy management algorithms, and next-generation battery technologies will further strengthen the system's performance and broaden its applicability.

This self-recharging hybrid propulsion system is an original conceptual idea developed by **Hind Meriem Mokadem**, one of the authors, at a secondary school level. It has been studied and developed based on research and initial engineering work and remains a promising concept that has not yet undergone full professional testing. The author plans to pursue it further through engineering studies in college to transform it into a functional, real-world sustainable marine propulsion system, a future in which boats no longer deplete the planet's resources but draw their power from the natural forces that surround them every day.

THEME 3: Global Trade Optimization: Post-1970 Policies and Future Propulsion Systems

3.1 Introduction

Humans have long depended on trade. Each trade route has generated capital, profit, and mutual exchange. In the 1970s, four major events directly affected global trade: rapid decolonization, improvements in maritime law and policy, the Cold War, and rapid growth in maritime technology, especially in fuel efficiency and sustainable energy. The evolution of post-1970 maritime policy and technological innovation, particularly containerization and fuel efficiency standards, significantly optimized global trade flows (UNCTAD, 2021).

This analysis will address the impact of policy and innovation on trade flows and the reshaping of maritime trade policies by the 1973 oil crisis. It will analyses several trade laws and policies while focusing on emerging marine propulsion technologies that align with conservation and trade efficiency goals.

3.2 The Rising Importance of Maritime Trade and Fossil Fuel Dominance

The world before the 1970s was a fast-paced place. After the Second World War, the relevance of coal as a global fuel declined. From the 1940s onward, the world went through two major events: decolonization and the Cold War. The transition of marine trade vessels from coal-powered engines to hydrocarbon engines was revolutionary. This transition later ignited the growth of the petroleum and fossil fuel industries around the world. The power held by Britain, France, the Netherlands, Spain, and Portugal diminished after the independence of the British Raj, Indochina, the Dutch East Indies, and other colonies. This power shift started in the early 1940s and lasted until the 1980s.

After World War II ended in 1945, the triad emerged: a group of three regions that dominated trade, namely North America, Western Europe, and the Asia-Pacific. In a world of rapid rebuilding, fossil fuels were a key ingredient for both the East and the West. Britain had lost its influence over the Gulf countries and South Asia, so securing oil became a personal interest. Routes like the Suez Canal became important to the West. Trade centers such as New Orleans, Rotterdam, and Liverpool became significant. In Asia, Mumbai, Aden, and Singapore showed promise as top-tier trade hubs (UNCTAD, 2023).

Transition from coal to hydrocarbons: Coal was still important for making steel and electricity, but oil took its place for fueling ships, trains, and homes.

Energy dependency: In the 1960s, industrial economies depended heavily on imported crude oil, which made them vulnerable to supply disruptions in the Middle East.

Infrastructure growth: During this period, specialized ships like oil tankers became more common.

Oil had all the features that were desirable for trade. It was easy to transport and had excellent fuel efficiency. It allowed automated fuelling, replacing the labour that coal needed. It was also cleaner and less hazardous than coal. This made the switch from coal to petroleum resources easier. Thus, oil and petroleum became important for transport and trade. Diesel produces 43–46 MJ/kg of energy output, compared to coal's 24 MJ/kg (Natrium; Fossil vs. Alternative Fuels).

The Use of Coal for Electricity Production

Table 3.1 – Share of coal in electricity production, France and the USA (1970–2010)

Country/Years	1970	1980	1990	2010
France	30.55%	27.35%	8.49%	approx. 4–5%
USA	46.39%	51.20%	53.07%	44.8%

Reading the table, the share of coal in French electricity production fell by about 22 percentage points between 1970 and 1990 (from 30.55% to 8.49%) and kept falling to roughly 4–5% by 2010. Over the same period, the share of coal in the USA rose from 46.39% in 1970 to 53.07% in 1990, before decreasing to 44.8% in 2010.

Why is the data above so conflicting? Why did coal usage skyrocket in the USA while France saw a major decline in coal usage?

These two conflicting factors can be connected to several historical events. The 1973 and 1979 oil crises played a major role. While natural gas, diesel, and petroleum are used in industries and boat engines, they are not primarily used for electricity generation. During the Cold War and the oil crisis, the USA cut down energy production by oil because it did not want a breach in energy

security. It switched back to coal, which had been quietly declining in influence in the energy-producing industries (EIA).

France, however, chose a different path. Instead of returning to coal, France industrialized and began commodifying nuclear energy for its electricity production. Under the Messmer Plan, France established nuclear energy as the primary electricity-generating source, reducing its dependency on fossil fuels (Asia Times; Wikipedia).

Back to oil. Oil has always been a concern of energy security and geopolitics. According to UNCTAD, 90% of world trade is carried out via oceans (UNCTAD). The fuel that ships use is HFO (heavy fuel oil). The importance of oil and petrol is enhanced by their growing role in geopolitics. The supply of petroleum, mineral oils, and other fuel resources usually passes through disputed, conflicted, or unstable areas, making it an extremely vulnerable resource. Major conflicts, such as the current geopolitical tensions between the West and the Gulf countries, show how oil can be a fatal friend to a country's economy.

3.3 Case Study: India

The legal framework regarding Indian marine laws and maritime activities is primarily centered on the Merchant Shipping Act of 1958. This serves as an umbrella for future laws and policies regarding maritime activities, especially shipping regulations and marine pollution (Indian Code). However, this law was heavily influenced by international laws, conventions, and policies that existed at the time, as well as ones that came into existence after them. There are a few key components of the law existing in the legal framework today:

Regulatory authority: The responsibilities of maintaining, enforcing, controlling, and protecting Port State Control (PSC) are in the hands of the Directorate General of Shipping (DG Shipping) and the Mercantile Marine Departments (MMD). They have the authority to conduct surveys and examinations of foreign vessels to check whether they follow international regulations such as MARPOL, SOLAS, and the International Safety Management Code.

Pollution and liability: Because of extreme globalization and to protect its international image, India's Merchant Shipping Act now follows several international regimes:

- MARPOL 73/78 (Part XI-A), which governs pollution from marine vessels, regulating routine and unintended releases.
- The Civil Liability Convention of 1992 (Part X-B), which states that shipowners shall handle any damage to marine bodies caused by oil pollution.
- The IOPC Fund 1992 (Part X-C), which was enforced to financially compensate individuals affected by oil spills.
- The Bunker Oil Convention (2001), which India ratified in 2015, establishing a framework for damages caused by oil spills from a ship's fuel and cargo-related incidents.

Fuel standards: India, by following MARPOL Annex VI, has established a 0.5% m/m limit on Sulphur content in fuel. This applies to all vessels in Indian territorial waters from January 2020.

Arrest and claims: The Admiralty Act of 2017 enables a legal framework for ship arrests regarding maritime claims, such as those arising from oil pollution damage. The Merchant Shipping (Amendment) Bill 2022 aims to enhance pollution classifications, add air pollutants and waste materials with new frameworks, and revise ship registration procedures.

Impact of oil on maritime conservation laws: Oil's catastrophic effects on marine ecosystems, including suffocating coral reefs and mangroves (like the Sundarbans) and causing mass fish and seabird deaths, have been a major factor in the development of India's environmental protection policies.

Frameworks for reaction: The National Oil Spill Disaster Contingency Plan (NOS-DCP) was established in 1993 because of the ecological hazard posed by oil. It gives the Indian Coast Guard the structure to organize quick responses to spills, even if it is not legally obligatory.

Judicial oversight: The National Green Tribunal (NGT) Act of 2010 established a prompt legal process for environmental disputes, including the payment of damages caused by marine pollution.

Strict responsibility standards: In response to decisions such as Rylands v. Fletcher, where the "polluter pays" principle is enforced regardless of fault, Indian courts and laws have shifted toward a strict responsibility paradigm for oil spills.

Impact of oil on trade policy: India has been compelled to include environmental compliance in its marine trade policy due to concerns associated with oil pollution and global energy trends.

Mandatory financial security: Before entering Indian ports, both foreign and Indian boats are now required to carry Certificates of Insurance or other financial security to cover civil responsibility for oil damage.

Economic protectionism: Policies that prioritize the financial recovery of local businesses through the IOPC fund and strict responsibility against shipowners result from spills that interrupt important trade sectors like fishing and coastal tourism.

Global compliance as a commerce barrier: The IMO 2020 Sulphur Cap, which forbids the transportation of non-compliant fuel unless a vessel is outfitted with exhaust gas cleaning systems (scrubbers), has significantly altered commerce. Although it standardizes competition, it raises merchants' operating expenses by ensuring that ships stopping at Indian ports adhere to international environmental norms.

3.4 Case Study: Indonesia

The main foundation for Indonesia's marine conservation and commerce initiatives is the Global Maritime Fulcrum (GMF) program formally established by Presidential Regulation No. 16 of

2017 on Indonesian Ocean Policy. With a focus on the oceanic economy, national security, and ecological preservation, this framework coordinates nautical activities across several departments.

Rules concerning Indonesian marine conservation

- The Fisheries Law (No. 31 of 2004) is the main piece of legislation for protecting natural ecosystems and aquatic life. Law No. 45 of 2009 strengthened enforcement measures by amending it, giving investigators the authority to demolish or submerge foreign vessels involved in illegal fishing.
- National conservation laws have recently been amended to include obligations regarding maritime zones. These revisions created harsher penalties for crimes against wildlife, with fines of up to 50 billion rupiah and jail terms of up to 20 years.
- Customary law communities, such as Pang lima Laot in Aceh and Awig-awig in Bali, have legal standing under the 1945 Constitution. In contrast to state-sponsored inspection, these communities provide social validation and customary punishments that often result in better adherence to resource preservation.

Economic and trade policies

- The Blue Economy and the elimination of Illegal, Unreported, and Unregulated (IUU) fishing are major focuses of Indonesia's marine commerce strategy.
- An estimated IDR 30 trillion in yearly economic losses are caused by IUU fishing. Due to trade policy, all commercial vessels must have particular permits, such as SIUP for business operations, SIPI for fishing operations, and SIKPI for transportation, to ensure traceability and adherence to environmental regulations.
- Over 3.3 million kg of ornamental fish were exported from Indonesia between 2015 and 2019, making it a significant global supplier in the marine pet sector. There is a tension between environmental sustainability and financial well-being because this commercial activity is mostly unregulated and relies on animals collected from their natural habitats.
- To protect domestic processing resources, the government might impose export limits on raw materials used in the fishing sector.

Impact of petroleum on Indonesia's legal system

The petroleum and natural gas industry has served as a catalyst for policy coherence within the Indonesian framework, but it also poses a significant environmental risk.

- The National Ocean Policy was primarily prompted by the growth of deep-sea and offshore oil and gas development. To reduce inter-sectoral disputes related to shipping, fishing, and energy extraction, the administration sought to abandon the "sectoral approach."

- **Environmental protection standards:** When it comes to the management of Marine Protected Areas, oil spills are seen as a major risk and hindrance. The national plan of action now includes "marine environment management" as a cluster of priority initiatives.
- **Maritime diplomacy:** Indonesia aggressively pursues regional collaboration to eradicate maritime conflict causes, particularly by tackling marine pollution. By combining Indonesia's goals with regional initiatives like the ASEAN Community and India's "Act East" strategy, the policy hopes to promote environmental security and regional stability.

3.5 The Issue of Weak Maritime Laws and Policies in Europe and Asia

Humans have had laws in place for different interactions for thousands of years. Whether for political systems or criminal offences, we have long had laws to regulate society. So why should we not also set some ground rules for our ships, which represent over 80% of the global trade of goods? Especially in the wake of the environmental crisis following the Industrial Revolution, maritime law has become increasingly important in reducing global emissions.

3.5.1 The MARPOL Convention

The International Convention for the Prevention of Pollution from Ships (MARPOL) is the main international convention covering the prevention of marine pollution by ships from operational or accidental causes, according to the International Maritime Organization. The first MARPOL Convention was adopted in November 1973.

MARPOL Annex VI, which entered into force in 2005, is an IMO treaty setting global limits on air pollutants from ship exhaust, including Sulphur oxides, nitrogen oxides, and ozone-depleting substances. It regulates fuel quality, shipboard incineration, and establishes stringent Emission Control Areas (ECAs) to protect health and the environment.

These regulations include:

- Prohibition of any deliberate emission of ozone-depleting substances.
- Prohibition of new installations containing ozone-depleting substances.
- Nitrogen oxide limits: 17.0 g/kWh when $n < 130$ rpm; $45.0 \times n^{(-0.2)}$ g/kWh when $130 \leq n < 2000$ rpm; 9.8 g/kWh when $n \geq 2000$ rpm.
- Sulphur oxide regulations: a limit of 4.5% Sulphur content of any fuel oil used on board ships. Within a SOx Emission Control Area, the Sulphur content must not exceed 1.5% m/m, or an approved exhaust gas cleaning system must be applied to reduce emissions to 6.0 g SOx/kWh or less.

These values are the original Annex VI limits, which have been tightened since then: the limit inside Emission Control Areas has been 0.10% m/m since 1 January 2015, and the global limit has been 0.50% m/m since 1 January 2020.

Have these regulations worked? According to the Report of the 2018 Concentrated Inspection Campaign (CIC) on MARPOL Annex VI by the Tokyo MOU, compliance is generally very good. During the CIC period, a total of 8,270 inspections were carried out involving 7,657 ships. Of these, 6,604 inspections were performed using the CIC questionnaire, and 198 ships (2.58%) were detained.

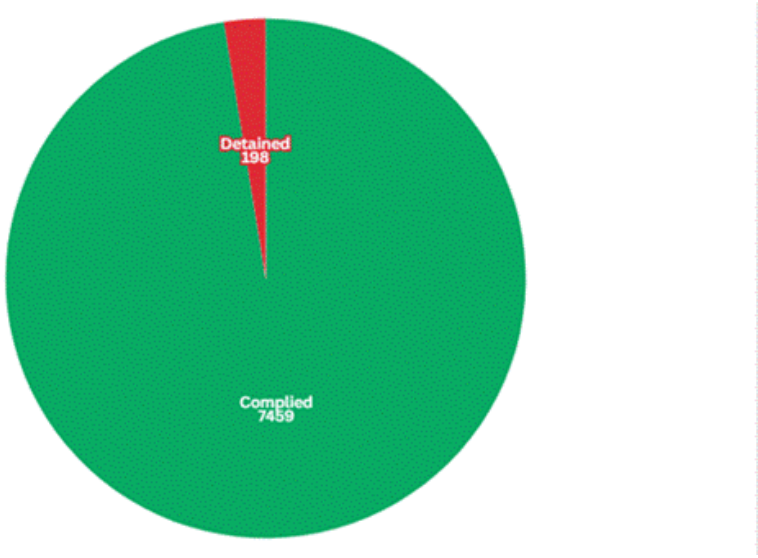


Figure 3.1 – Outcome of the 2018 Concentrated Inspection Campaign on MARPOL Annex VI: ships that complied and ships detained. Based on data from Tokyo MOU (2018).

Figure 3.1 shows that, out of the 7,657 ships concerned, 7,459 complied and 198 were detained.

The best compliance was on Q2: "Do bunker delivery notes indicate that fuel oils delivered and used on board do not exceed the maximum allowed Sulphur content?" with 99.98% answering "yes." The worst compliance was on Q6, concerning the maintenance of the ozone-depleting substances record book, with only 94.84% answering "yes."

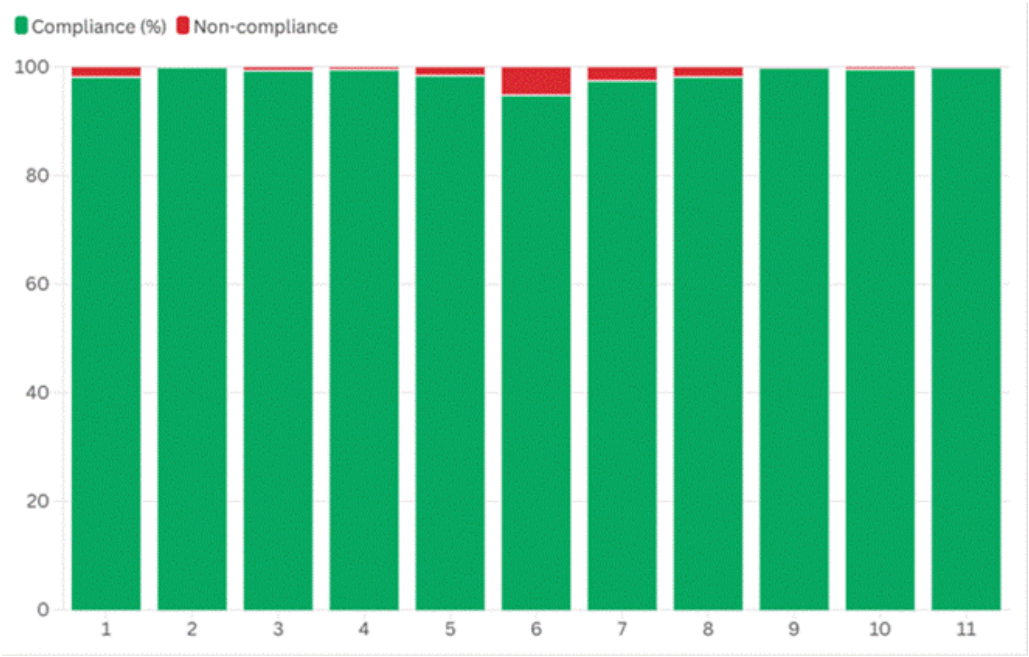


Figure 3.2 – Compliance and non-compliance (%) for each of the 11 questions of the 2018 CIC questionnaire. Based on data from Tokyo MOU (2018).

Figure 3.2 shows, for each of the 11 questions, the percentage of ships that complied and the percentage that did not. In almost every case the non-compliance share is barely visible, and Question 6 shows the widest gap. For Question 11, "no" is the correct answer.

According to the "Evaluation of the response to emerging environmental threats, focusing on carbon dioxide (CO2), volatile organic compounds (VOCs), and scrubber wash water (SOx)" published in the Euro-Mediterranean Journal for Environmental Integration, "the existing international regulatory system is more efficient today than in past years."

In conclusion, the regulations set out in MARPOL Annex VI have helped reduce emissions.

3.5.2 The United Nations and Its Efforts in Trade Efficiency and Environmental Conservation

The United Nations has long recognised that the growth of international maritime trade cannot come at the expense of the world's oceans and atmosphere. Through three of its most influential bodies — the International Maritime Organization (IMO), the United Nations Conference on Trade and Development (UNCTAD), and the United Nations Environment Programme (UNEP) — the UN has built a comprehensive framework that seeks to govern global shipping on two fronts: ensuring that trade moves efficiently, and that it does so without causing irreversible environmental harm.

The UN's institutional framework

The IMO is the body responsible for enforcing MARPOL regulations and designating Emission Control Areas, making it the most direct instrument the UN has for managing the environmental impact of global trade. It focuses on maritime safety, environmental preservation, and fair-trade conduct. The IMO Annual Report 2024 shows that this framework is actively producing reform: new Emission Control Areas, strengthened marine litter regulations, and a clear timeline for future environmental designations.

UNCTAD approaches maritime governance from a trade and development perspective. Rather than regulating ships directly, it monitors the economics of maritime transport, tracks how shipping policies affect developing nations, and publishes its annual Review of Maritime Transport.

UNEP addresses the environmental dimension of maritime governance, providing scientific assessments and supporting international cooperation on marine pollution and ecosystem protection.

Emission Control Areas and the fight against air pollution

The IMO Annual Report 2024 laid out several significant reforms. One of the most notable developments was the designation of the Mediterranean Sea as an Emission Control Area for Sulphur Oxides under MARPOL Annex VI, which came into effect on 1 May 2024. This designation limits the sulphur content in fuel used by ships in the area to 0.10%, a dramatic reduction aimed at cutting air pollution in one of the world's busiest shipping corridors. Looking ahead, the Canadian Arctic Waters and the Norwegian Sea are set to become Emission Control Areas from March 2027.

It should be noted that the amendment creating the Mediterranean ECA entered into force on 1 May 2024, while the 0.10% sulphur limit itself applies to ships from 1 May 2025.

The impact of these measures is already visible in emissions data. Reports from the European Maritime Safety Agency indicate large decreases in sulphur oxide emissions, particularly following the tightening of fuel sulphur limits. During the COVID-19 pandemic, SOx emissions fell by as much as 70%. Significantly, rather than rebounding as global trade recovered, emissions have stabilised at this lower level and continued to decline, suggesting that stricter regulations, once implemented, produce lasting environmental benefits. On this basis, it is recommended that additional regions be added to the list of Emission Control Areas.

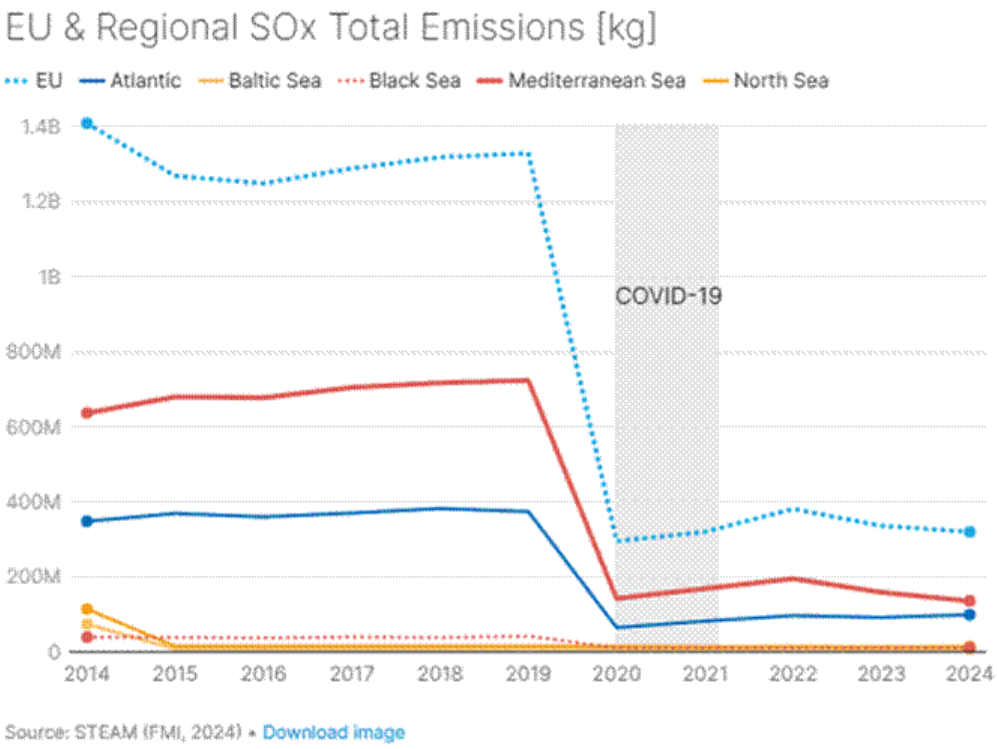


Figure 3.3 – EU and regional SOx total emissions (kg), 2014–2024. Source: STEAM (FMI, 2024).

Figure 3.3 confirms this trend. Read from the chart, total SOx emissions in the EU fall from about 1.3 billion kg in 2019 to about 0.3 billion kg in 2020, and the Mediterranean Sea falls from about 0.7 billion kg to about 0.15 billion kg over the same period. The fall coincides with the entry into force of the global 0.50% Sulphur limit on 1 January 2020 and with the COVID-19 period shaded on the chart. Emissions then remain far below their pre-2020 level until 2024, which supports the idea that the reduction is lasting.

3.6 Emerging Marine Propulsion Technologies

The regulations described in this theme push the shipping industry away from heavy fuel oil. In 2023, the IMO adopted a revised strategy on greenhouse gas emissions from ships, which aims at net-zero emissions by or around 2050, with indicative checkpoints in 2030 and 2040 (IMO, 2023). Reaching these targets requires changes in both the fuels used on board and the propulsion technologies that burn or convert them.

Cleaner conventional engines: Liquefied natural gas (LNG) dual-fuel engines almost eliminate sulphur oxide emissions and reduce nitrogen oxides and carbon dioxide compared with heavy fuel oil. Their climate benefit is limited, however, by methane slip, which is the release of unburned methane. Scrubbers, mentioned in the India case study, allow ships to keep burning high-sulphur fuel while meeting sulphur limits.

Alternative fuels: Methanol is liquid at ambient temperature and relatively easy to handle, but it carries roughly half the energy per kilogram of conventional marine fuel. Ammonia contains no carbon, but it is toxic and its combustion must be controlled to limit nitrogen oxides. Hydrogen has a very high energy content per kilogram, but it needs bulky storage, either compressed or at cryogenic temperatures, which reduces cargo space.

Electric and hybrid propulsion: Batteries are already practical for short routes such as ferries. For longer voyages, hybrid diesel-electric systems combine engines and batteries so that engines can run at an efficient load and batteries cover peaks and port operations. Connection to shore power in ports further reduces emissions at berth.

Wind-assist systems: Rotor sails, rigid sails, and kites use wind to supply part of the propulsion power. The fuel savings depend on the route and weather, so these systems complement the main engine rather than replace it.

Table 3.2 – Approximate energy content and main trade-offs of marine energy carriers

Energy carrier	Approx. energy content (MJ/kg)	Main advantage	Main limitation
Heavy fuel oil / diesel	40–46	Mature technology and global bunkering network	Sulphur, nitrogen oxide and CO2 emissions
LNG	approx. 50	Almost no Sulphur oxides; lower CO2	Methane slip; cryogenic storage
Methanol	approx. 20	Liquid at ambient temperature	Low energy density, needs larger tanks
Ammonia	approx. 19	No carbon in the fuel	Toxicity; nitrogen oxide control
Hydrogen	approx. 120	Highest energy per kilogram; no CO2 at the point of use	Very low energy per volume; bulky storage

No single option is ideal. Energy density affects the range of a ship and the space left for cargo, so each technology involves a trade-off between emissions, cost and trade efficiency. This echoes the lesson of the 1973 oil crisis discussed in Theme 4: depending on a single energy source makes trade vulnerable, and diversification is a strategy for both energy security and environmental protection.

THEME 4: The 1973 Oil Crisis and Its Aftereffects

4.1 Introduction to Trade Crises and the 1973 Oil Crisis

The 1973–74 oil crisis stands as one of the most dramatic economic shocks in the post-war world, but its roots reach deeper than the eruption of war in the Middle East. For years, oil-producing nations had been pushed to take more power from the Western oil giants that dominated global markets. The Arab Israeli War in October 1973 simply gave these producers a chance to flex new-found leverage.

The shock forced governments around the world to rethink their approach to energy. Oil was no longer just a critical fuel for transport, industry, and power. Now, governments worried about their reliance on foreign oil, supply security, and the fallout if shipments dried up without warning. The effect was immediate and far-reaching, rattling international politics, upending trade flows, fueling inflation, and forcing countries to overhaul their energy policies.

The Yom Kippur War, which erupted on October 6, 1973, saw Egypt and Syria launch a coordinated attack to reclaim territories lost to Israel in 1967. As the fighting escalated, so did the involvement of Cold War superpowers. The U.S. airlifted military aid to Israel under Operation Nickel Grass, while the Soviet Union shipped arms to Egypt and Syria. American support for Israel triggered a decisive move from Arab oil producers. On October 17, 1973, the Arab members of OAPEC announced steep production cuts and targeted embargoes against states backing Israel, with the United States and the Netherlands among the main targets.

This embargo did not emerge out of nowhere. Throughout the early 1970s, OPEC had gained muscle in pricing talks, able to dictate terms that were once set by Western oil majors. The impact of their new power was stunning: oil prices shot from about $3 a barrel to as much as $12 by early 1974. The IMF later described this as leading to a tripling, while some accounts call it a fourfold jump, all depending on what numbers are used.

This was not just a matter of long queues at gas stations. Modern economies, completely dependent on oil for transportation, manufacturing, and even food production, felt the sting everywhere. The price shock rippled across industries, and households found themselves paying much more for everything from heating to groceries.

The crisis unfolded at the height of the Cold War, adding geopolitical edge to an already volatile situation. The oil producers' ability to cut exports and push up prices revealed new power balances. No longer were the U.S. and Soviet Union the only major players. Oil producers had a resource everyone else needed, and they were willing to use it.

Even so, Middle Eastern states did not achieve total control overnight. Western oil companies still had enormous technical expertise and marketing clout. Over the next several years, however, producing governments ramped up their ownership and oversight of domestic oil industries, a process that extended well beyond 1973.

4.2 Economic Consequences

Economic troubles came swiftly. The sudden leap in energy costs hammered industrialized economies. Producers paid more for fuel, transport firms struggled with soaring prices, and households everywhere felt new strains. Growth rates lagged, but inflation soared, a combination that economists began calling "stagflation." Countries highly dependent on imported energy, like Japan, were hit especially hard.

Developing nations faced their own disaster. They had weaker financial reserves, so the price hikes strained their economies even more severely. An IMF estimate for 1974 lays the disparity bare: oil exporters recorded an enormous current-account surplus of roughly $68 billion, while industrialized nations and developing oil importers faced deficits of $31 and $34 billion respectively. In effect, money was flowing out of consumer countries and into the coffers of oil producers.

With so much cash piling up in oil-exporting states, the question became: what next? The oil-rich countries could not spend all their new wealth at home. Instead, they stashed large sums in Western banks, which then recycled the "petrodollars," lending them to cash-strapped governments and businesses elsewhere. This recycling kept the system afloat, but spurred fresh rounds of borrowing, particularly in developing countries, notably across Latin America. When interest rates spiked in the early 1980s, debts became unsustainable, laying the groundwork for the developing world debt crisis.

4.3 Policy Responses and Strategic Reactions

On the policy front, governments everywhere scrambled to adjust. Energy policies that once revolved around cheap, plentiful imports had to be rewritten. Countries asked hard questions: Who supplies our oil? How vulnerable are we if supplies are cut off? Can we do with less, or tap into domestic alternatives? The responses were as diverse as the countries confronting the crisis.

One major collective step was the creation of the International Energy Agency (IEA) in 1974. The IEA aimed to foster cooperation among oil-importing countries, improve energy security, and organise a coordinated response to future crises. In the U.S., the Energy Policy and Conservation Act of 1975 created the Strategic Petroleum Reserve, America's emergency oil stash, stored in underground caverns along the Gulf Coast.

The IEA was created in November 1974 within the framework of the OECD, and the same 1975 American law that created the Strategic Petroleum Reserve also introduced the fuel economy standards described below.

Governments also tried to rein in demand. The U.S. introduced Corporate Average Fuel Economy (CAFE) standards, forcing automakers to make their fleets more efficient. American consumers started looking for smaller, more fuel-efficient cars, creating new opportunities for Japanese manufacturers already ahead of the curve. Fuel economy transformed from a technical detail to a major selling point.

Another shift: countries began to look for oil beyond traditional Middle Eastern fields. High prices made new ventures worthwhile. Production spread to places like the North Sea, Alaska, and Mexico, while offshore drilling surged. Diversification reduced dependence on just a handful of suppliers, changing the competitive landscape of the oil market.

4.4 Impact on France

France did not just tweak its energy policy after the oil shocks; it went all-in on nuclear power. Before 1973, France leaned hard on imported oil and other energy from abroad. So, when oil prices shot up, it turned into a real headache for the country. The answer, at least for the French government, was to cut back on fossil fuel imports by building nuclear power plants.

In March 1974, Prime Minister Pierre Messmer unveiled a massive nuclear program. The plan centered on pressurized-water reactor (PWR) technology, and they wanted those reactors to run fast. The slogan was "all-electric, all-nuclear." Electricity was shifting to come almost entirely from nuclear power, and that meant France could ditch a lot of the oil it used to buy for electricity.

Still, the oil crisis was not the birth of the French nuclear dream. Nuclear research and development were already rolling before 1973; France had scientists and engineers who knew their stuff. The crisis just hit the accelerator, scaling up the government's nuclear ambitions.

France also rolled out energy-saving rules. In 1974, the government started cutting back on things like building lighting, advertising signs, and road fuel use. They set up the Agence pour les Économies d'Énergie to push conservation too. These choices stuck. By the 1980s and 1990s, nuclear power delivered most of France's electricity. That shift meant France needed a whole lot less imported oil for its energy than it did back in the early seventies.

4.5 Impact on India

India felt the crisis in a totally different way. Oil imports drained the economy. Homegrown energy was already in short supply, so when oil prices spiked, India's foreign exchange reserves took a hit.

An IMF study showed that the petroleum import bill in 1973 was about 40 percent of the value of all India's commodity exports. The World Bank noted that the bill nearly tripled in 1973 and doubled again by 1974. The country suddenly needed a lot more foreign currency to keep those oil shipments coming.

That pain motivated the government to dig for domestic sources and step up its hunt for energy. Bombay High was a gamechanger. India ramped up oil production off its western coast. Between the start of the 1970s and 1979–80, Bombay High went from producing nothing to delivering around 36% of India's domestic crude. So, France went nuclear, but India drilled for oil. Both chased energy security but went about it in ways that fit their resources and economies.

4.6 Impact on Indonesia

Indonesia's story is different. As an OPEC member and oil producer, it was technically sitting pretty, at least on paper. Since joining OPEC in 1962, Indonesia could, in theory, cash in on the higher prices. But things were not so simple. Indonesia exported oil but had a big, growing population and serious development needs. They needed to buy machines from abroad, build infrastructure, and keep oil powering their own economy.

So, not all those oil profits stuck around. A lot got soaked up by spending on roads and schools, and imports. Saudi Arabia, Kuwait, and the UAE, oil-rich countries with small populations, made out like bandits. Indonesia? It had more mouths to feed and a longer to-do list. It is too easy to say "all OPEC countries got rich." It really depended on production numbers, population size, how much oil the home crowd needed, and how wisely governments spent and saved.

But Indonesia did benefit from the bigger shift: petroleum producers started wrestling control back from multinational oil companies in the 1970s. Ownership and revenue debates all flared up, with more power shifting to the country that actually owned the oil.

4.7 The Petroleum Industry in Flux

Take Indonesia's oil story, its roots go back to colonial times when Royal Dutch Petroleum built up operations. By the 1970s, producers all over the world wanted a bigger cut. That meant renegotiating contracts, nationalizing assets, or setting up deals where the state got more say and more profits. Big oil companies, the Seven Sisters, did not vanish, but their power to dictate terms started to fade a bit. This did not happen overnight, but the oil crisis of 1973 made those negotiations much tougher for the companies.

4.8 Maritime and Environmental Ripple Effects

New drilling technology pushed oil exploration off the coast and into the sea. It opened up fresh reserves but brought fresh risks too: tanker spills, offshore accidents, and damage to fisheries and marine life. For Indonesia, a nation built around islands and the sea, these issues hit close to home. Oil, fishing, shipping, and environmental concerns all competed for space and attention. It is not fair to say that the 1973 crisis directly sparked all of Indonesia's modern maritime policies, but the race for offshore oil definitely raised new problems about how to juggle energy, shipping, fisheries, and environmental protection.

4.9 Debt Troubles in the Developing World

Oil-importing countries suddenly needed more cash to keep the energy flowing. Meanwhile, banks in Europe and the US were flooded with petrodollars from oil exporters. They started lending, and borrowing shot up in much of the developing world.

At first, this did not look too risky: interest rates stayed low and global trade had momentum. But cracks started showing by the end of the 1970s. US interest rates spiked under Paul Volcker's

Federal Reserve, and those countries with variable-rate loans saw their payments balloon quickly. Then came the second oil shock around 1979–80. Oil-importing countries paid more, economies slowed down, and servicing those loans became a struggle.

By the early 1980s, some governments were borrowing just to pay the bills on their old debts. Mexico's announcement in 1982 that it could not pay up marked the breaking point for developing-world finances. Sure, oil was not the only reason: too much borrowing, bad policies, fallen commodity prices, and the recession all fed into the crisis. Still, the oil shocks and the trail of recycled petrodollars helped set the stage.

4.10 Why All This Mattered

The 1973 oil crisis threw the world's priorities into chaos. Rich countries realised those long oil supply chains could snap or get squeezed. Suddenly it was all about fuel efficiency, new sources, and emergency preparedness. Oil exporters found out that control over energy could be political power, not just a cash cow. For developing countries stuck importing oil, the pressure on their economies was intense. Some went deep into debt just to keep the lights on.

Some governments, like France, made lasting changes, building nuclear plants at a record pace. The US set up the Strategic Petroleum Reserve. India dug for homegrown petroleum. The International Energy Agency was born to coordinate among oil importers. The relationship between oil companies and producing countries also changed. Producer governments bargained harder, rewrote contracts, and took more control.

4.11 Conclusion

The oil crisis of 1973 started with war but ended up shaking the whole world. What began between Egypt, Syria, and Israel spun into a bigger power game with America and the Soviet Union on opposite sides. Oil prices soared, and everyone, factories, drivers, farmers, had to pay up.

Oil money poured into the world's oil capitals, and a big chunk flowed back into world finance as petrodollars. That fueled loans to developing countries, and when interest rates eventually jumped, trouble brewed.

Everywhere, governments adopted new energy strategies. The US set up an oil reserve. France built nuclear reactors like never before. India tried to develop domestic oil. The North Sea, Alaska, and Mexico brought in new producers. Oil grew more powerful politically: producer countries called more of the shots, energy policy was always on the table, and the big oil firms had to adjust.

So, the first oil shock was much more than a price spike at the pump. It tied together war, energy, finance, development, and international politics, and its leg

Theme 5:Proposed Legal Instruments for a Renewable Maritime Transition

The previous sections have examined the ecological damage caused by conventional marine engines (Theme 1), the development of an innovative concept of a selfrecharging hybrid propulsion system that can eliminate these impacts (Theme 2), the evolution of global maritime policy and trade (Theme 3), and the strategic vulnerability exposed by the 1973 oil crisis (Theme 4). These four perspectives converge on a single conclusion: the transition to renewable marine propulsion is not only a technological challenge but also a legal and political one. ***The following proposed instruments are intended to bridge that gap.*** They are not adopted legislation but structured legal frameworks designed to accelerate the adoption of clean propulsion technologies while aligning maritime trade with marine conservation.

The protocols below combine regulatory pressure, market incentives, and financial support. They target the environmental failures identified in Theme 1, directly promote the renewable system developed in Theme 2, build on the trade and policy frameworks analyzed in Theme 3, and draw strategic lessons from the energy crisis studied in Theme 4. In this way, they transform the historical and technical analysis of this research into a concrete policy pathway toward a more sustainable maritime future.

➢ Protocol 1 – Maritime Fleet Energy Sovereignty Protocol (MFESP)

Drawing on the lessons of the 1973 oil crisis, all motorized vessels operating in marine and ocean waters shall progressively reduce their reliance on petroleum-based propulsion. Each state shall define a national transition trajectory aligned with the objectives of the International Maritime Organization, adapted to its energy and economic situation. To support this transition, a Maritime Energy Sovereignty Fund shall be created, financed by a small levy on imported marine fuel and by international climate finance mechanisms. This fund shall subsidize the installation and maintenance of renewable marine propulsion systems, with priority given to coastal communities and small island states.

Why this instrument matters

The 1973 oil crisis showed that dependence on imported petroleum is a strategic and economic vulnerability. Existing regulations focus on large ships, leaving many motorized vessels unaddressed. This instrument fills that gap by requiring a transition in all marine and ocean waters, while the creation of a dedicated fund makes the transition financially feasible. It links the historical lesson of energy vulnerability with the development of renewable marine propulsion.

➢ **Protocol 2 – Marine Engine Noise Liability Protocol (MENLP)**

Any marine engine model exceeding the underwater noise emission standard shall be subject to a progressive noise pollution tax. The tax rate shall increase with the level of underwater noise produced. Vessels equipped with certified quiet propulsion systems, such as electric or renewable hybrid drives, shall be exempt from this tax.

Why this instrument matters

Underwater noise is a largely overlooked form of marine pollution. It disrupts communication, reproduction, and migration in marine mammals and fish, yet no binding international standard exists for small vessels. Existing IMO guidelines are voluntary. This instrument introduces the first progressive noise tax, creating a direct economic incentive to reduce engine noise without imposing abrupt bans. It directly benefits silent propulsion systems like the self-recharging hybrid developed in this study.

➢ **Protocol 3 – Clean Port Access Protocol (CPAP)**

Access to certified green ports shall be reserved exclusively for vessels powered by zero-emission or renewable propulsion systems. Vessels using fossil fuel engines shall be denied entry, except during a temporary transition period for those undergoing certified retrofits. Green ports shall be equipped with renewable energy infrastructure, such as solar panels and wind turbines, which supply clean electricity to dedicated charging stations for electric and hybrid vessels. These ports shall offer exclusive advantages to clean vessels, including priority berthing, reduced port fees, and access to these renewable charging facilities. Conventional ports remain open to all vessels without such benefits.

Why this instrument matters

Existing green port initiatives rely on voluntary schemes or discounts, but they do not restrict access. This instrument transforms ports into true clean-propulsion gateways. By making clean energy a condition of entry, it creates a powerful incentive for operators to transition. The integration of renewable infrastructure at the port itself also makes the port more autonomous and resilient, echoing the energy-sovereignty lesson of the 1973 crisis. The self-recharging hybrid system developed in this study would automatically qualify for full access and all exclusive benefits.

➢ Protocol 4 – Protected Coastal Zone Clean Access Protocol (PCZCAP)

Access to Marine Protected Areas and designated coastal fishing zones shall be progressively restricted to fishing vessels certified as clean and quiet. These vessels must use renewable or zero-emission propulsion systems that do not release fuel or oil into the water and do not exceed established underwater noise limits. The transition period and certification requirements shall be defined by each coastal state according to its national capacity and the maturity of available technologies. During this transition, small-scale fishers shall receive active support from the State to adapt their vessels so that they are not excluded or penalized and can continue to access their traditional fishing grounds.

Why this instrument matters

This instrument connects marine conservation with clean propulsion while ensuring social protection for small-scale fishers. Instead of relying on a simple deadline, it places responsibility on the state to actively support fishers during the transition. The self-recharging hybrid system automatically meets the clean and quiet requirements, making it a practical option for those who depend on protected coastal areas.

➢ Protocol 5 – Renewable Propulsion Reinvestment Fund Protocol (RPRFP)

All financial penalties and sanctions imposed on vessels for marine pollution violations, including fuel leaks, emissions, and underwater noise, shall be directed into a dedicated international fund. This fund shall be used exclusively to finance research, development, and deployment of renewable marine propulsion technologies. It shall support the broader maritime energy transition by accelerating innovation in clean propulsion systems, including hydrokinetic and solar-hybrid solutions. The fund shall be managed transparently, with regular public reporting to ensure that the resources collected from pollution are effectively reinvested in the transition to a sustainable maritime sector.

Why this instrument matters

Most pollution fines disappear into general state budgets and do not contribute to solving the problem that caused them. This instrument creates a direct link between environmental damage and its remedy. Every penalty paid by a polluting vessel becomes a resource for financing research and innovation in renewable marine propulsion. Instead of funding isolated projects, the fund accelerates the entire maritime energy transition, making clean technologies more mature and accessible. It transforms punishment into progress at the scale of the whole sector.

➢ **Protocol 6 – Vessel Emission Transparency Protocol (VETP)**

All motorized vessels shall display a mandatory standardized emission label indicating their CO_2 output, underwater noise level, and fuel type. This label must be registered in a public national database accessible to port authorities, coastal communities, and environmental agencies. The label shall be updated at each periodic inspection or when the vessel's propulsion system is modified. This requirement applies to all vessels, without exception, and does not impose any tax or restriction by itself.

Why this instrument matters

Transparency is the first step toward accountability. Large commercial ships already have mandatory efficiency certificates, but small vessels remain largely undocumented. This instrument fills that gap by making pollution data visible and accessible. It does not impose any penalty or restriction, but it creates the informational foundation that allows port authorities, buyers, and regulators to make informed decisions. A standardized label also encourages voluntary improvement: once emissions and noise are publicly visible, market pressure naturally favors cleaner and quieter vessels. In this way, this instrument supports the other proposed protocols without duplicating their mechanisms.

Results / Key Findings:

The research produced five main findings, each supported by the analysis developed in the corresponding theme. Taken together, they show that the transition to renewable marine propulsion is technically feasible, environmentally necessary, and politically achievable.

1. Conventional marine engines are actively destructive, not merely inefficient.

The analysis in Theme 1 showed that two-stroke, four-stroke, and small diesel engines release unburned hydrocarbons, nitrogen oxides, carbon monoxide, particulate matter, heavy metals, noise, and waste heat directly into coastal waters. These pollutants do not disperse harmlessly: they accumulate in sediments, contaminate the food chain, contribute to hypoxia, and disrupt the feeding and reproductive cycles of marine organisms. The damage is not a future risk; it is already measurable in coastal ecosystems. This finding establishes the environmental urgency that drives the rest of the research.

2. Renewable marine propulsion is technically feasible for small vessels.

Theme 2 demonstrated, through engineering calculations, that a self-recharging hybrid propulsion system combining solar photovoltaic panels and retractable hydrokinetic turbines can power a small coastal vessel without fuel or shore-side charging. At a current of 2 m/s, two MERIDYNE-70 turbines deliver more than 1 kW of continuous electrical power, while the solar array produces approximately 10.9 kWh per day. Together, these sources cover about 67% of the vessel's daily energy needs at sea, with the battery supplying the remaining deficit. The battery alone provides more than six days of full autonomy. This finding confirms that the system is not a theoretical concept but a dimensioned and calculated engineering proposal.

3. The MERIDYNE turbine fills a gap that no existing turbine addresses.

Existing hydrokinetic turbines are designed for tidal farms and cannot fit on small hulls. The MERIDYNE family, in contrast, is modular, retractable, and sized specifically for small coastal vessels. It deploys only when the boat is stationary, capturing energy from the natural current without interfering with propulsion, and retracts fully during navigation to eliminate drag. This design choice is what makes the entire system physically credible and operationally realistic. Without it, the concept would remain abstract. This finding positions MERIDYNE as the central innovation of the research.

4. Existing maritime governance leaves small vessels unaddressed.

The analysis in Themes 3 and 4 showed that current international frameworks, such as MARPOL, the EEDI, and the SEEMP, focus almost exclusively on large commercial ships. Small coastal vessels, which collectively consume enormous amounts of fuel, remain largely unregulated. The 1973 oil crisis revealed the strategic vulnerability of fossil-fuel dependence, yet the same vulnerability persists today for millions of small boats. This regulatory gap is a structural weakness that no purely technological solution can overcome. This finding justifies the need for a legal and political response.

5. A legal framework is needed to complete the transition.

The six protocols proposed in Theme 5 provide a coherent pathway. They combine regulatory pressure (noise tax, emission standards), financial incentives (reinvestment fund, port access advantages), conservation measures (protected coastal zones), and transparency mechanisms (emission labels). Together, they link the historical lesson of the 1973 oil crisis to the practical needs of today's coastal communities, and they make the transition fair, enforceable, and economically viable. This finding shows that engineering innovation and legal transformation must advance together.

FINAL CONCLUSION :

The objective of this research was to examine how the transition from fossil-fuel marine propulsion to renewable systems can be achieved while protecting marine ecosystems, preserving the livelihoods of coastal communities, and maintaining the stability of global maritime trade. To answer this question, the study followed a clear path: it identified the environmental damage caused by conventional engines, proposed an original renewable propulsion system, analyzed the historical and political context, and finally translated these findings into concrete policy instruments.

The conclusions are as follows. First, the ecological analysis confirmed that conventional marine engines are not merely inefficient; they are actively destructive. Their emissions, fuel leaks, noise, and waste heat disrupt marine life and contaminate the food chain. This damage is not a distant threat but a present reality for coastal ecosystems. Second, the engineering work demonstrated that a self-recharging hybrid propulsion system, combining solar energy and hydrokinetic energy through the MERIDYNE turbine, is technically feasible. It offers a realistic pathway toward autonomous, zero-emission operation for small vessels, even though it remains a conceptual innovation that requires further development.

Third, the historical and policy analysis showed that technology alone cannot succeed. The 1973 oil crisis proved that energy dependence is a strategic vulnerability, and the evolution of maritime governance has left significant gaps, particularly for small-scale fishers who are often ignored by international regulations. Fourth, the protocols and legal instruments proposed in Theme 5 go beyond filling gaps. They represent a necessary step toward a new maritime governance framework. By linking the lessons of the 1973 oil crisis to the development of renewable propulsion, they show that the transition cannot be left only to technological progress. It must be supported by clear, fair, and enforceable mechanisms that involve states, ports, manufacturers, and coastal users. These instruments are not simple recommendations; they are concrete policy pathways designed to embed renewable propulsion into the future of maritime trade and conservation.

In conclusion, this research demonstrates that marine conservation and maritime decarbonization are not separate goals. They require both engineering innovation and legal transformation. The self-recharging hybrid system is a seed for future development, and the proposed protocols are a framework for action. The next step must be deeper engineering studies, pilot projects, and international cooperation. The ocean is a shared resource, and its future depends on the choices

made now. This research offers one clear direction: move beyond fossil fuels, support renewable propulsion, and build the governance needed to make that transition fair and lasting.

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• Figure 2.3.5.a—SunPower Maxeon 400 W monocrystalline panel:
Source: SunPower / Maxeon (image: empowersolaraustralia.com.au).

• Figure 2.3.5.b—$LiFePO_4$ marine battery:
Source: Victron Energy (https://www.victronenergy.com).

• Figure 2.3.5.c — Marine PMSM motor:
Source: Bellmarine (www.bellmarine.com), a transfluid company.

• Figure 2.3.5.d — Marine electrical cabinet:
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